MIL-M-28787/267B
20 January 1984
SUPERSEDING
MIL-M-28787/267A(EC)
18 September 1981

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

BUFFER, THREE-STATE, DIGITAL, KEY CODE YBZ

* This specification is approved for use by all Departments and Agencies of the Department of Defense.

SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic three-state buffer module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.0462 failures per million operating hours for class I or 0.0652 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be YBZ.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/267-1 for class I and M28787/267-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

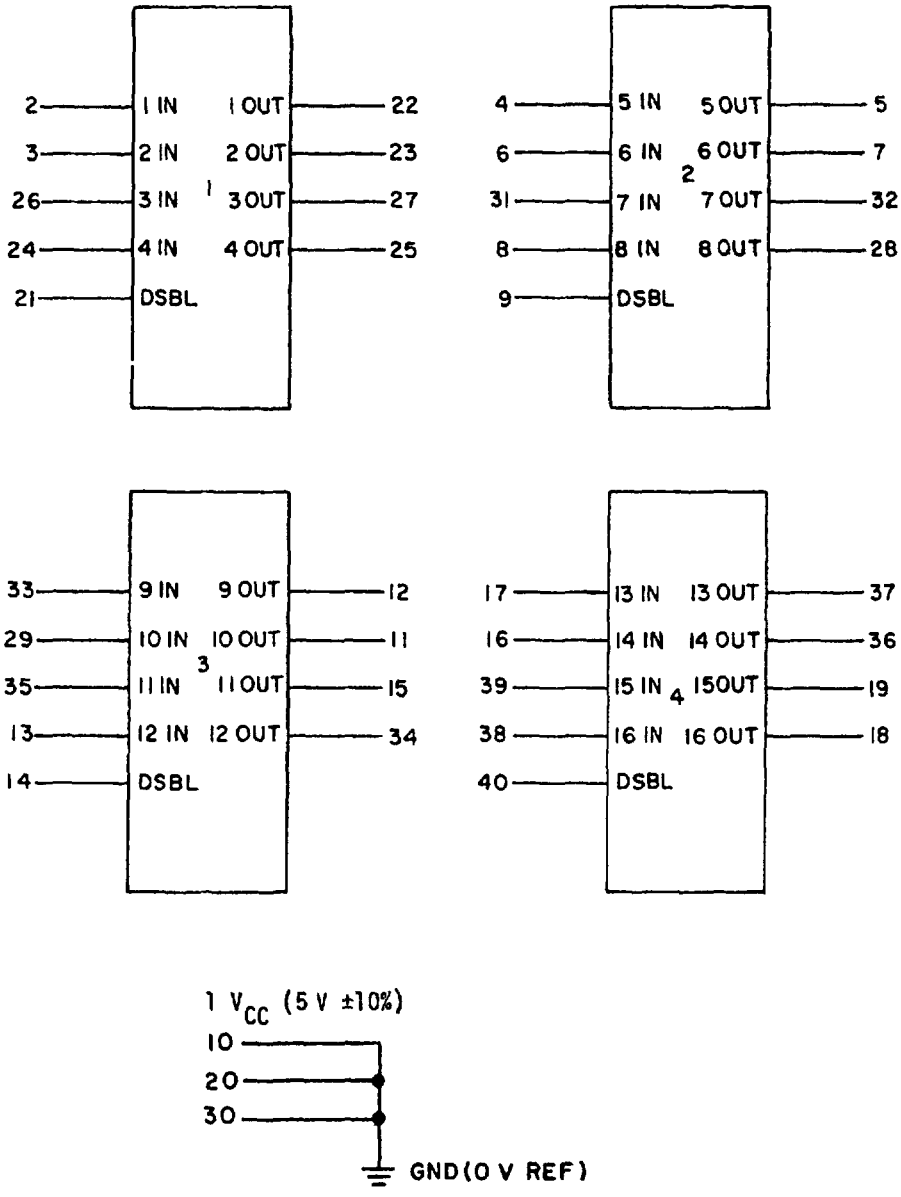
4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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* FIGURE 1. YBZ module function diagram, three-state buffer.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B16	Maximum stress current check	IV	S	← See table IV →										
C1-C16	V_{OUT} (low loaded)	V	S	10	800	10	800	10	800	10	800	10	800	mV
D1-D16	Input breakdown current	VI	S	0	1	0	1	0	1	0	1	0	1	mA
D17-D20	Input breakdown current, disable	VI	S	0	4	0	4	0	4	0	4	0	4	mA
E1-E16	Output short circuit current	VII	X	-30	-70	-30	-70	-30	-70	-30	-70	-30	-70	mA
F1-F20	Input clamp diode voltage	VIII	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
G1-G16	Output clamp diode voltage	VIII	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
H1-H16	Output leakage current, high level	IX	X	-40	40	-40	40	-40	40	-40	40	-40	40	µA
J1-J16	Output leakage current, low level	X	X	-40	40	-40	40	-40	40	-40	40	-40	40	µA
K1-K16	Output voltage, low level	XI	X	0	400	0	400	0	400	0	400	0	400	mV
L1-L16	Output voltage, high level	XI	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
M1-M16	Input current, high level	XII	X	0	40	0	40	0	40	0	40	0	40	µA
M17-M20	Input current, high level	XII	X	0	160	0	160	0	160	0	160	0	160	µA
N1-N16	Input current, low level	XIII	X	-0.10	-1.73	-0.10	-1.73	-0.10	-1.73	-0.10	-1.73	-0.10	-1.73	mA
N17-N20	Input current, low level	XIII	X	-0.10	-6.92	-0.10	-6.92	-0.10	-6.92	-0.10	-6.92	-0.10	-6.92	mA

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
P1	Power supply current, high level	XIV	X	50	216	50	216	50	216	50	216	50	216	mA
P2	Power supply current, low level	XIV	X	50	248	50	248	50	248	50	248	50	248	mA
Q1-Q16	Transition time (t_{TLH}), C_L = 50 pF	XV	S	4	25	4	25	4	28	4	25	4	30	ns
R1-R16	Transition time (t_{THL}), C_L = 50 pF	XV	S	4	20	4	20	4	23	4	20	4	25	ns
S1-S16	Propagation delay time (t_{PLH}), C_L = 50 pF	XVI	S	5	15	5	18	5	18	5	21	5	21	ns
S17-S32	Propagation delay time (t_{PLH}), C_L = 1000 pF	XVI	Q	10	50	10	55	10	55	10	60	10	60	ns
T1-T16	Propagation delay time (t_{PHL}), C_L = 50 pF	XVI	S	5	18	5	21	5	21	5	24	5	24	ns
T17-T32	Propagation delay time (t_{PHL}), C_L = 1000 pF	XVI	Q	10	50	10	55	10	55	10	60	10	60	ns
U1-U16	Disable delay time (t_{OH}), C_L = 50 pF	XVII	S	5	20	5	20	5	25	5	20	5	30	ns
V1-V16	Disable delay time (t_{HO}), C_L = 50 pF	XVII	S	5	25	5	25	5	30	5	25	5	35	ns
W1-W16	Disable delay time (t_{1H}), C_L = 50 pF	XVII	S	5	25	5	25	5	30	5	25	5	35	ns
X1-X16	Disable delay time (t_{H1}), C_L = 50 pF	XVII	S	5	23	5	23	5	28	5	23	5	31	ns
Y1-Y16	Output crosstalk, high level	XVIII	Q	0	400	0	400	0	400	0	400	0	400	mV
Z1-Z16	Output crosstalk, low level	XVIII	Q	0	1000	0	1000	0	1000	0	1000	0	1000	mV
AA	Power supply transient amplitude	XIX	Q	0	500	0	500	0	500	0	500	0	500	mV

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
BB	Power supply filter capacitance	XX	S	1.6	2.8	1.6	2.8	1.6	2.8	1.6	2.8	1.6	2.8	µF
CC1-CC16	Input capacitance	XX	Q	2	15	2	15	2	15	2	15	2	15	pF
CC17-CC20	Input capacitance	XX	Q	8	60	8	60	8	60	8	60	8	60	pF
DD1-DD16	Output capacitance	XX	Q	4	30	4	30	4	30	4	30	4	30	pF

1/ "X" parameters shall be 100 percent tested at 25°C during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent, or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

* 4.4.2 Order of testing. Test conditions A through DD shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

* 4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through DD in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through BB shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

* 4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column and test condition BB shall be performed using the initial limits at 25°C ±5°C ambient.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests E through BB shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests E through DD shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

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* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Units
Maximum power dissipation (outputs not loaded)	1.36	1.36	W
Input voltage			
Low level	-0.5 min, 0.8 max	-0.5 min, 0.8 max	V
High level	2 min, V_{CC} max	2 min, V_{CC} max	V
Input current (for all inputs except disable input)			
Low level (V_{IL} = 0.0 V)	-1.73 max	-1.73 max	mA
(V_{IL} = 0.4 V)	-1.6 max	-1.6 max	mA
High level (V_{IH} = 2.4 V)	40 max	40 max	µA
(V_{IH} = 5.5 V)	1 max	1 max	mA
Input current (disable input)			
Low level (V_{IL} = 0.0 V)	-6.92 max	-6.92 max	mA
(V_{IL} = 0.4 V)	-6.4 max	-6.4 max	mA
High level (V_{IH} = 2.4 V)	160 max	160 max	µA
(V_{IH} = 5.5 V)	4 max	4 max	mA
Output drive current maximum capability			
Low level (V_{OL} = 0.4 V)	16	16	mA
High level (V_{OH} = 2.4 V)	-2	-2	mA
Output leakage current			
V_{OL} = 0.4 V	-40 min, 40 max	-40 min, 40 max	µA
V_{OH} = 2.4 V	-40 min, 40 max	-40 min, 40 max	µA

6.5 Module function. This module consists of 4 four input three-state line drivers. Each driver operates in a single ended driver mode or a high impedance output mode.

a. To operate as 16 single ended line drivers, apply low level to the output disable inputs (to keep the outputs in the normal low impedance mode). All channels operate with no signal inversion.

b. When operating in a bus-oriented system with outputs tied directly to outputs of similar devices, all devices except one shall be placed in the high-impedance state. This is accomplished by insuring that a high level is applied to the output disable inputs of all devices except the one that is used to drive the bus. The driving device provides good capacitive drive capability and waveform integrity especially on the positive transition.

6.5.1 Input to output relationship. Each buffer has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship. 1/ - 3/

Inputs		Outputs
Disable	Data in	Data out
L	L	L
L	H	H
H	X	High impedance

1/ H = High level.
2/ L = Low level.
3/ X = Don't care.

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6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

6.6 Cross reference of classification. Military Specification MIL-M-28787/267 supersedes Department of the Navy, Naval Electronic Systems Command Drawing No. 28687-0102-822. Part number M28787/267-1 supersedes part number 0102-822.

6.7 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - ER
Navy - EC
Air Force - 85

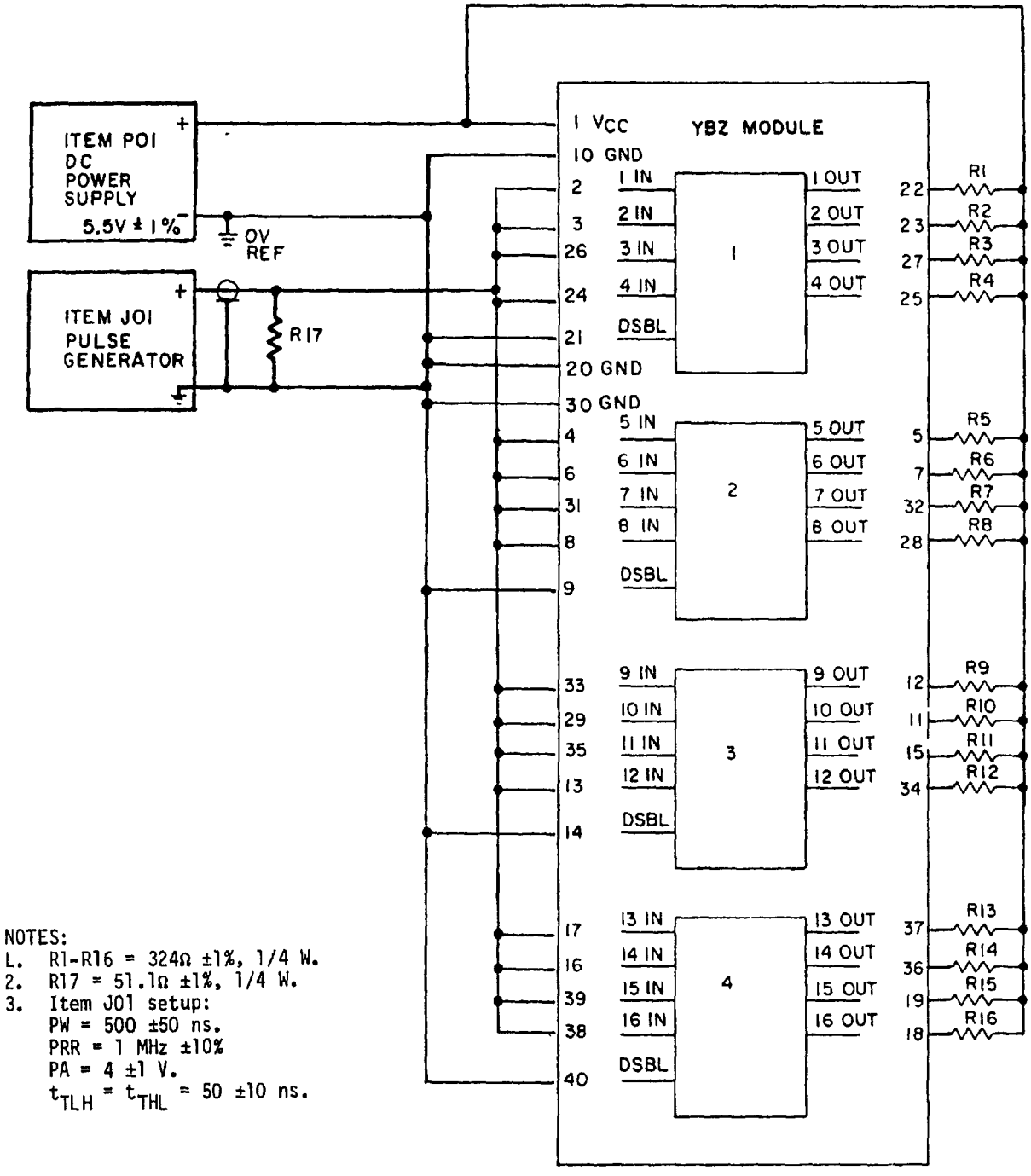
Review activities:
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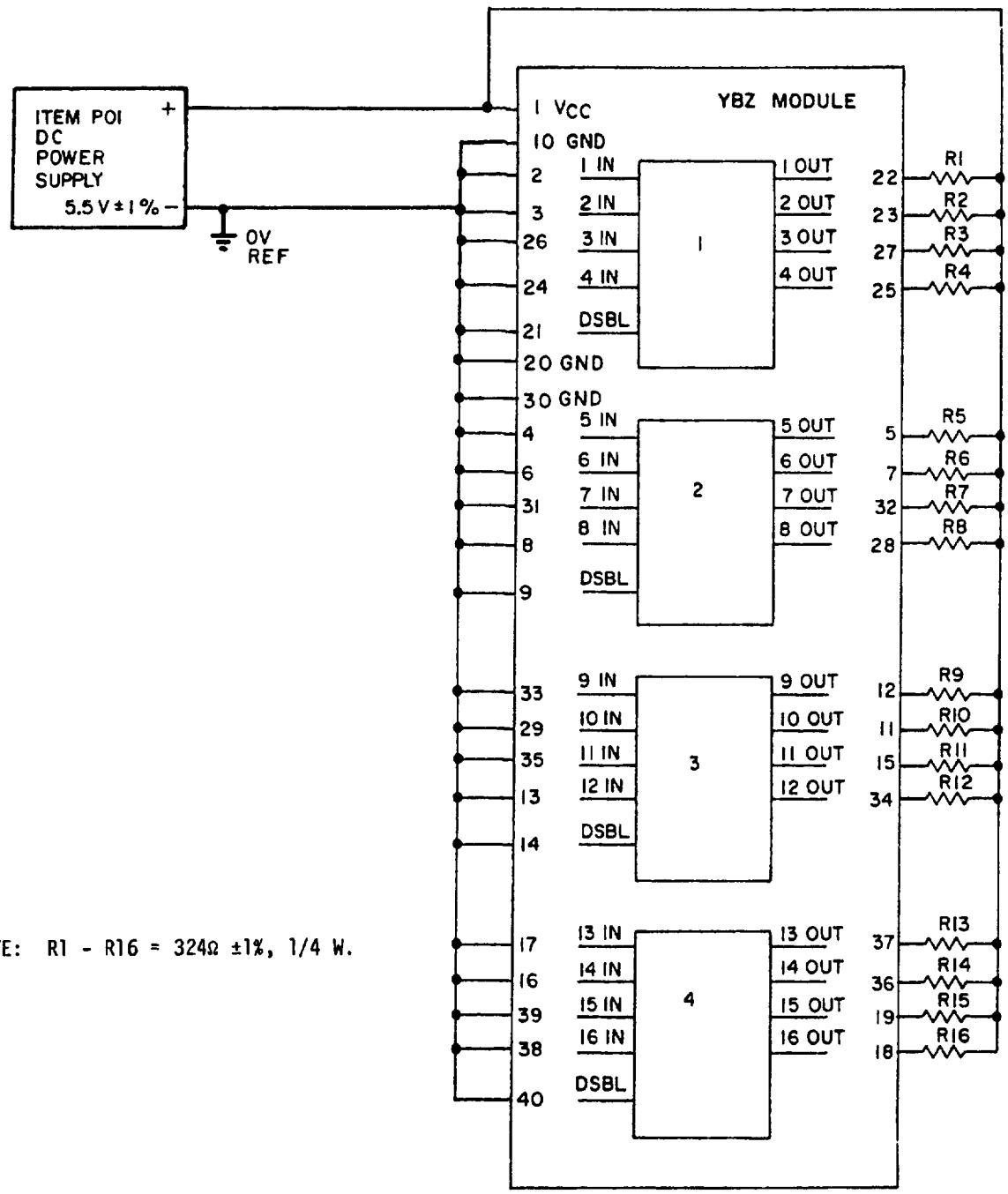
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NOTES:
1. R1-R16 = 324Ω ±1%, 1/4 W.
2. R17 = 51.1Ω ±1%, 1/4 W.
3. Item J01 setup:
   PW = 500 ±50 ns.
   PRR = 1 MHz ±10%
   PA = 4 ±1 V.
   $t_{TLH} = t_{THL}$ = 50 ±10 ns.

* FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1 - R16 = 324Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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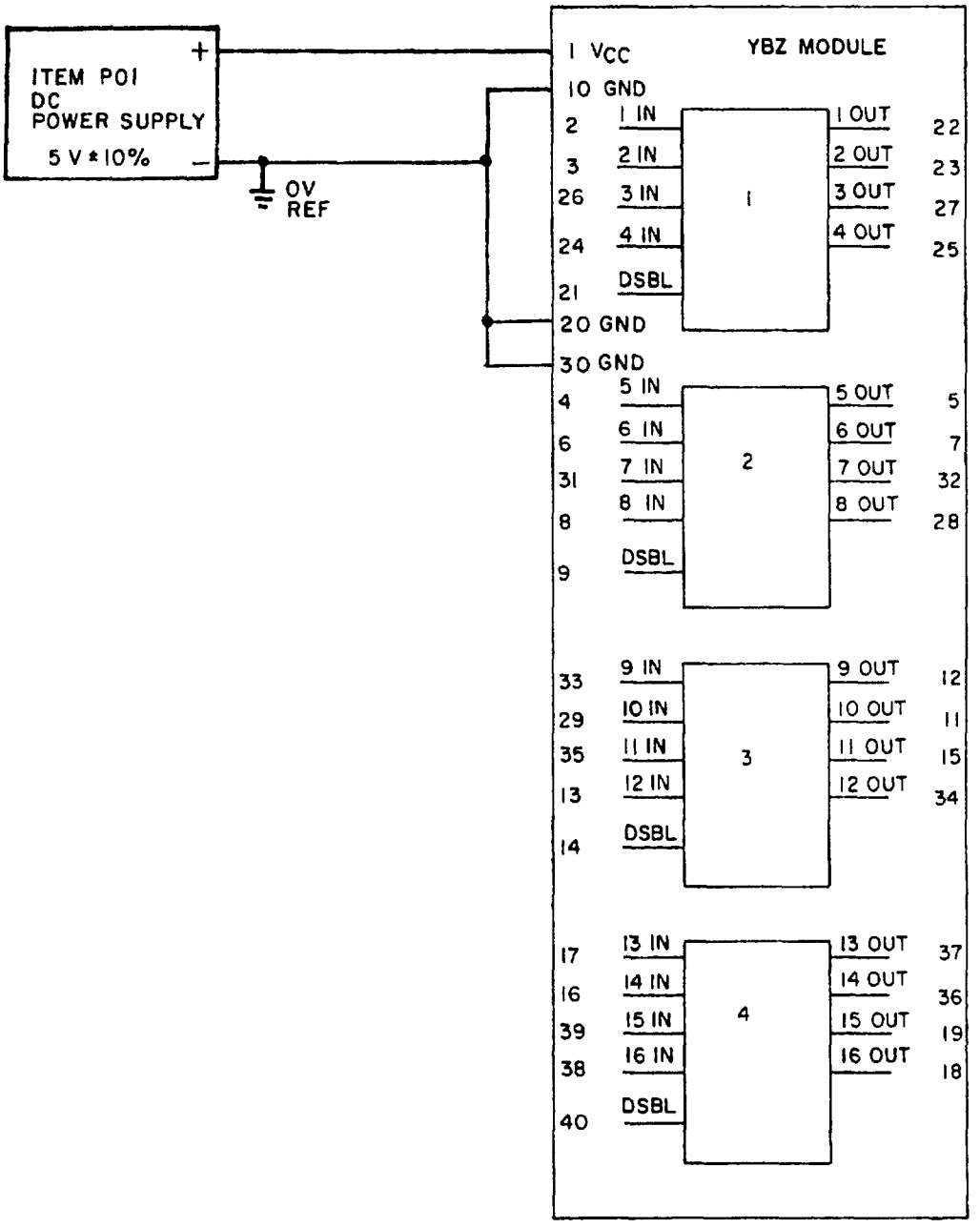


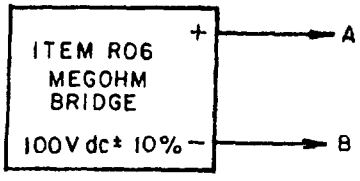
FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation, frame to module pins	All pins	Frame	10	∞	MΩ	5

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WARNING.

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative lead.

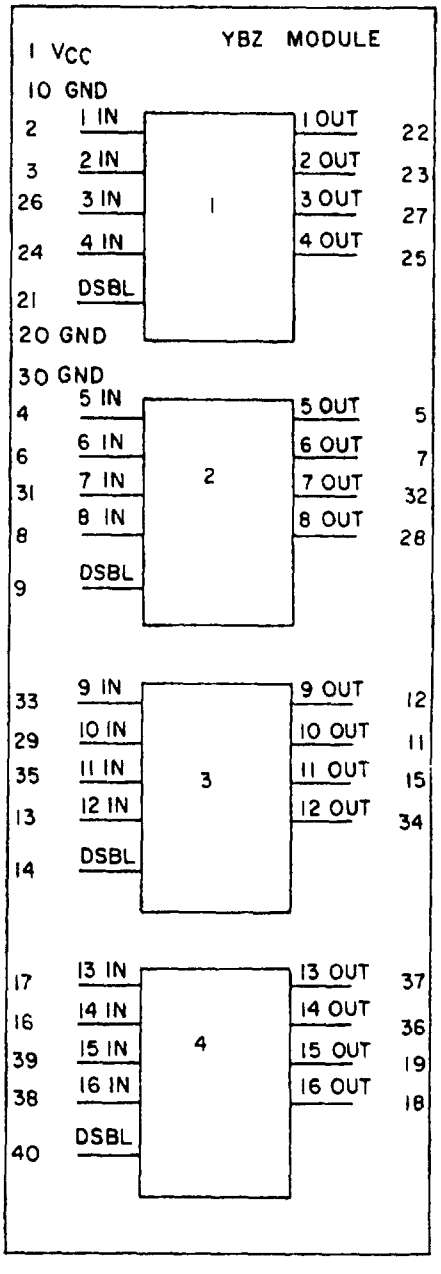


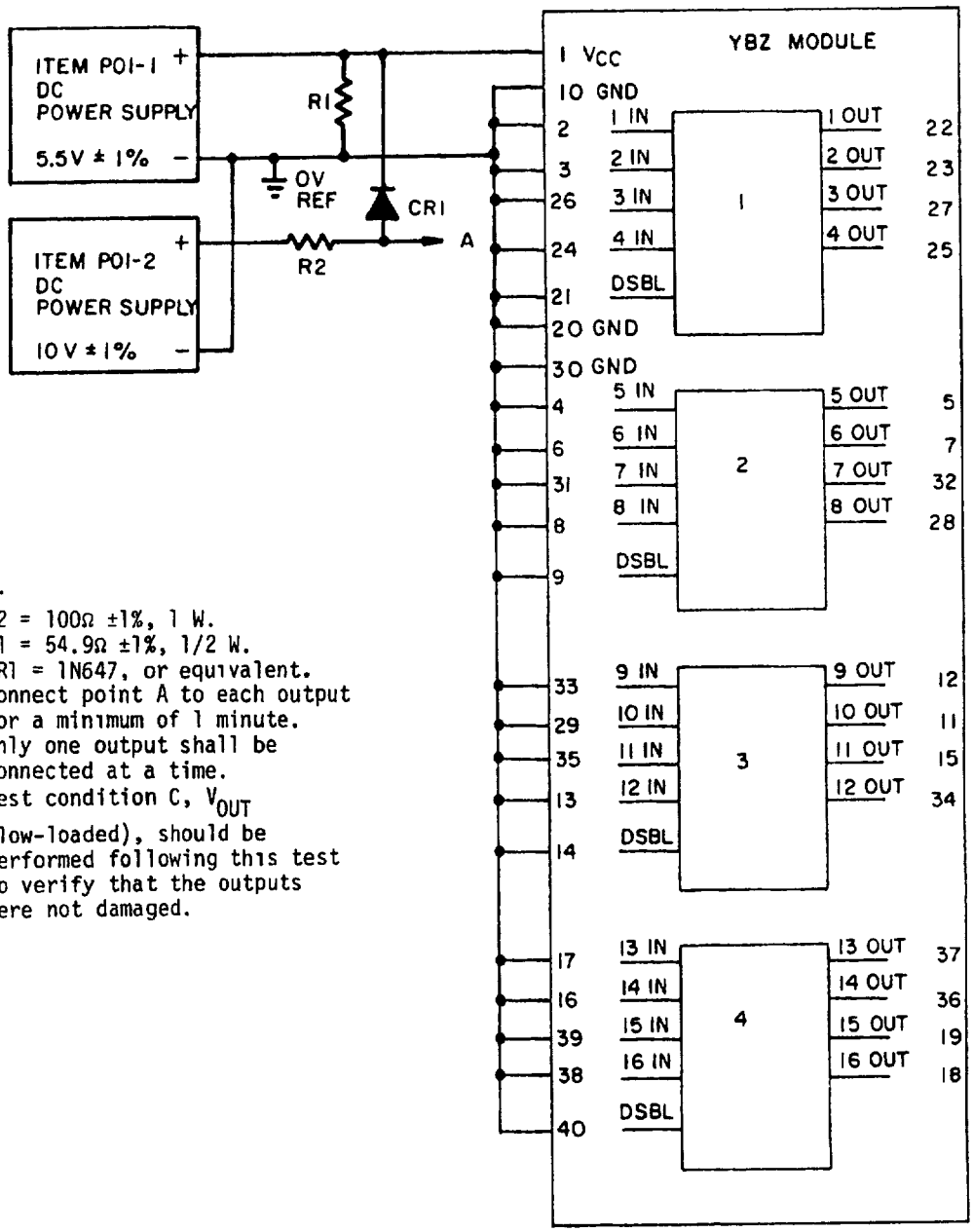
FIGURE 5. Isolation test circuit for table III.

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TABLE IV. Maximum stress current check test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Test figure
B1	1 Out Maximum stress current	22	See note 6 of figure 6		
B2	2 " "	23	"		"
B3	3 " "	27	"		"
B4	4 " "	25	"		"
B5	5 " "	5	"		"
B6	6 " "	7	"		"
B7	7 " "	32	"		"
B8	8 " "	28	"		"
B9	9 " "	12	"		"
B10	10 " "	11	"		"
B11	11 " "	15	"		"
B12	12 " "	34	"		"
B13	13 " "	37	"		"
B14	14 " "	36	"		"
B15	15 " "	19	"		"
B16	16 " "	18	"		"

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NOTES:
1. R2 = 100Ω ±1%, 1 W.
2. R1 = 54.9Ω ±1%, 1/2 W.
3. CR1 = 1N647, or equivalent.
4. Connect point A to each output for a minimum of 1 minute.
5. Only one output shall be connected at a time.
6. Test condition C, V_{OUT} (low-loaded), should be performed following this test to verify that the outputs were not damaged.

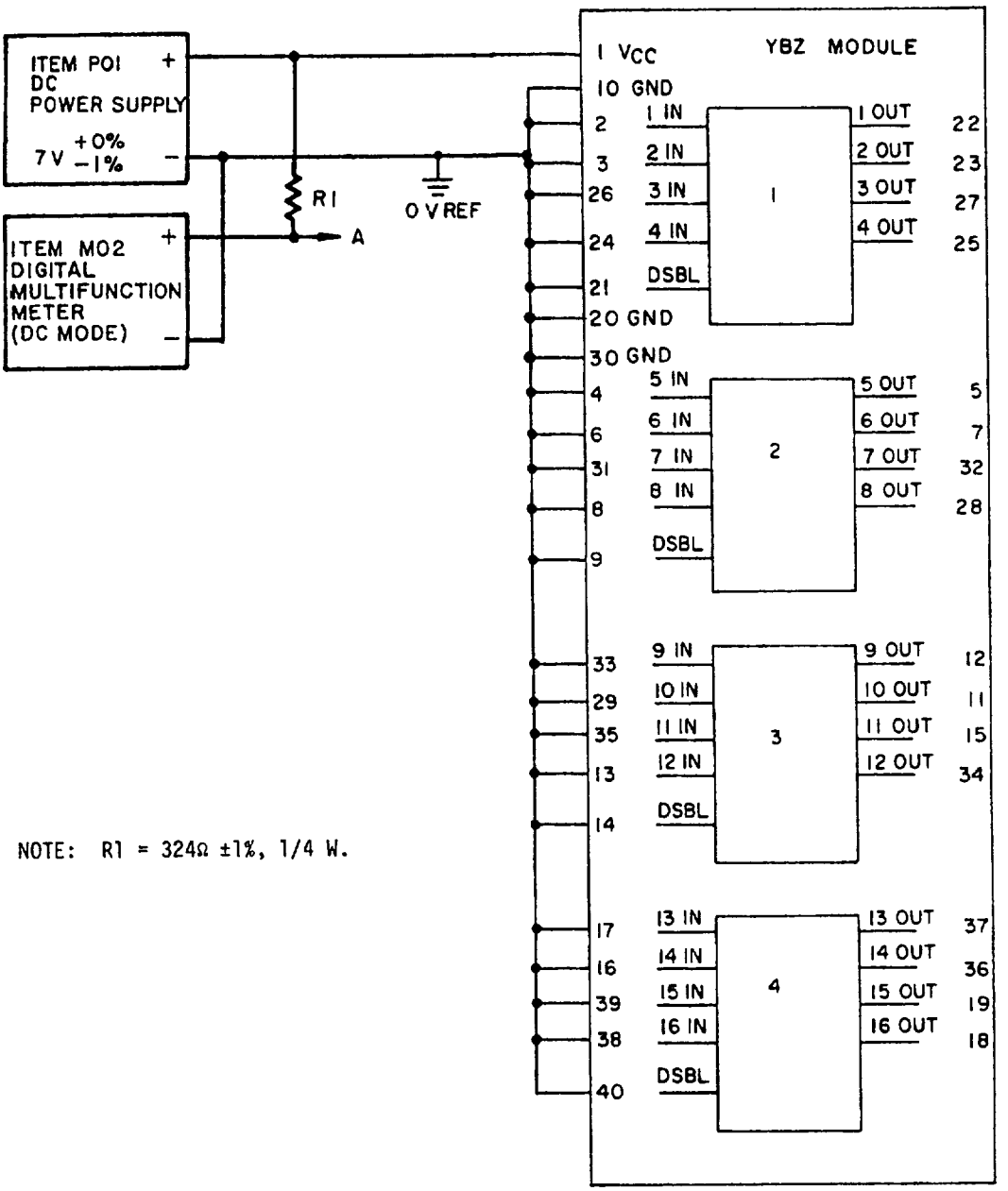
FIGURE 6. Maximum stress current check test circuit for table IV.

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TABLE V. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
C1	1 Out V_{OUT} (low loaded)	22	10	800	mV	7
C2	2 " "	23	"	"	"	"
C3	3 " "	27	"	"	"	"
C4	4 " "	25	"	"	"	"
C5	5 " "	5	"	"	"	"
C6	6 " "	7	"	"	"	"
C7	7 " "	32	"	"	"	"
C8	8 " "	28	"	"	"	"
C9	9 " "	12	"	"	"	"
C10	10 " "	11	"	"	"	"
C11	11 " "	15	"	"	"	"
C12	12 " "	34	"	"	"	"
C13	13 " "	37	"	"	"	"
C14	14 " "	36	"	"	"	"
C15	15 " "	19	"	"	"	"
C16	16 " "	18	"	"	"	"

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NOTE: R1 = 324Ω ±1%, 1/4 W.

*FIGURE 7. V_{OUT} (low-loaded) test circuit for table V.

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TABLE VI. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	1 Input breakdown current	3,26,24,4,6,31,8,33,29,35,13,17,16,39,38,21,9,14,40	2	0	1	mA	8
D2	2 "	2,26,24,4,6,31,8,33,29,35,13,17,16,39,38,21,9,14,40	3	"	"	"	"
D3	3 "	2,3,24,4,6,31,8,33,29,35,13,17,16,39,38,21,9,14,40	26	"	"	"	"
D4	4 "	2,3,26,4,6,31,8,33,29,35,13,17,16,39,38,21,9,14,40	24	"	"	"	"
D5	5 "	2,3,26,24,6,31,8,33,29,35,13,17,16,39,38,21,9,14,40	4	"	"	"	"
D6	6 "	2,3,26,24,4,31,8,33,29,35,13,17,16,39,38,21,9,14,40	6	"	"	"	"
D7	7 "	2,3,26,24,4,6,8,33,29,35,13,17,16,39,38,21,9,14,40	31	"	"	"	"
D8	8 "	2,3,26,24,4,6,31,33,29,35,13,17,16,39,38,21,9,14,40	8	"	"	"	"

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TABLE VI. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D9	9 Input breakdown current	2,3,26,24,4,6,31,8,29,35,13,17,16,39,38,21,9,14,40	33	0	1	mA	8
D10	10 "	2,3,26,24,4,6,31,8,33,35,13,17,16,39,38,21,9,14,40	29	"	"	"	"
D11	11 "	2,3,26,24,4,6,31,8,33,29,13,17,16,39,38,21,9,14,40	35	"	"	"	"
D12	12 "	2,3,26,24,4,6,31,8,33,29,35,17,16,39,38,21,9,14,40	13	"	"	"	"
D13	13 "	2,3,26,24,4,6,31,8,33,29,35,13,16,39,38,21,9,14,40	17	"	"	"	"
D14	14 "	2,3,26,24,4,6,31,8,33,29,35,13,17,39,38,21,9,14,40	16	"	"	"	"
D15	15 "	2,3,26,24,4,6,31,8,33,29,35,13,17,16,38,21,9,14,40	39	"	"	"	"
D16	16 "	2,3,26,24,4,6,31,8,33,29,35,13,17,16,39,21,9,14,40	38	"	"		"

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TABLE VI. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D17	1 Input breakdown current, disable	2,3,26,24,4,6,31,8,33,29,35,13,17,16,39,38,9,14,40	21	0	4	mA	8
D18	2 "	2,3,26,24,4,6,31,8,33,29,35,13,17,16,39,38,21,14,40	9	"	"	"	"
D19	3 "	2,3,26,24,4,6,31,8,33,29,35,13,17,16,39,38,21,9,40	14	"	"	"	"
D20	4 "	2,3,26,24,4,6,31,8,33,29,35,13,17,16,39,38,21,9,14	40	"	"	"	"

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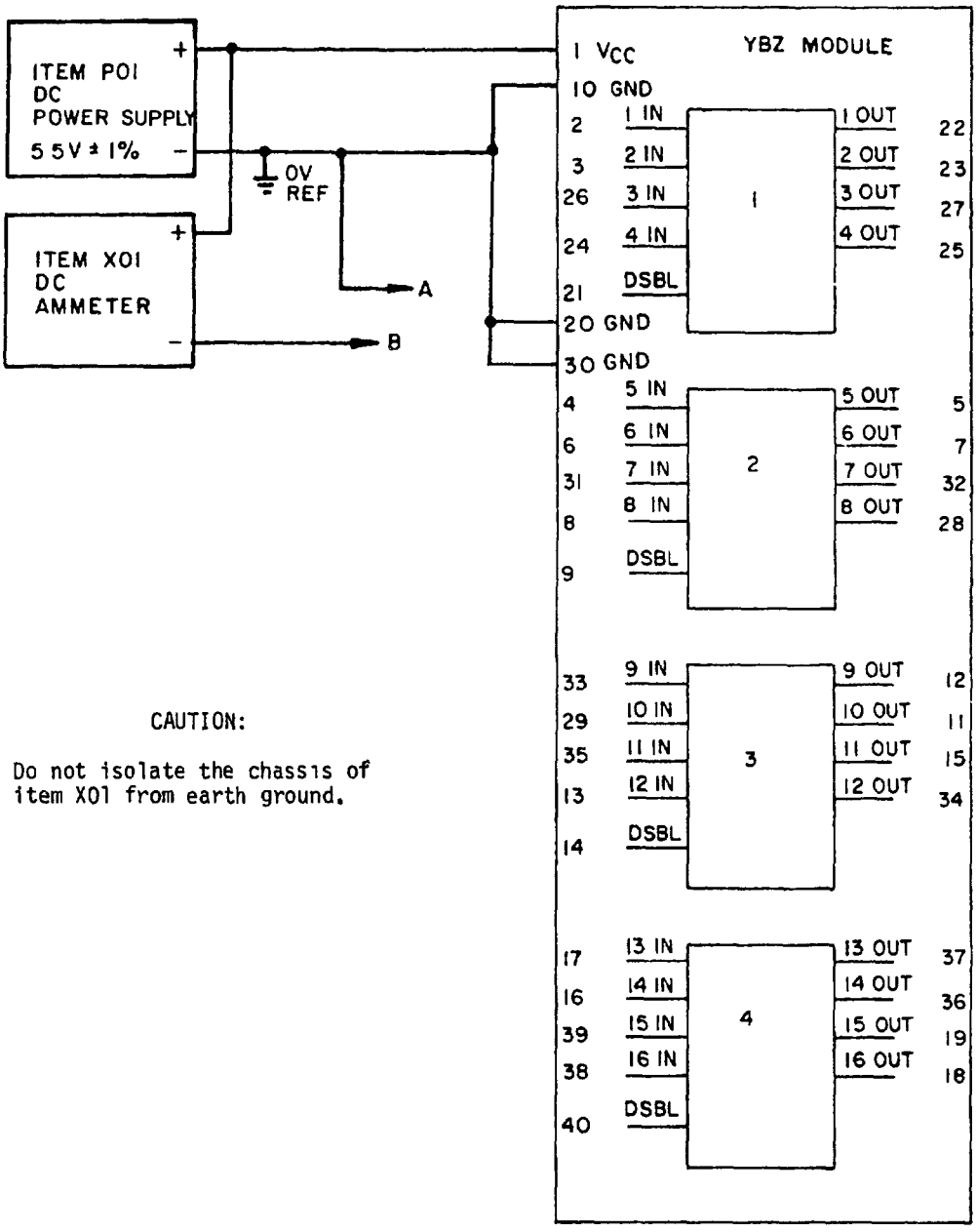


FIGURE 8. Input breakdown current test circuit for table VI.

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TABLE VII. Output short circuit current test.

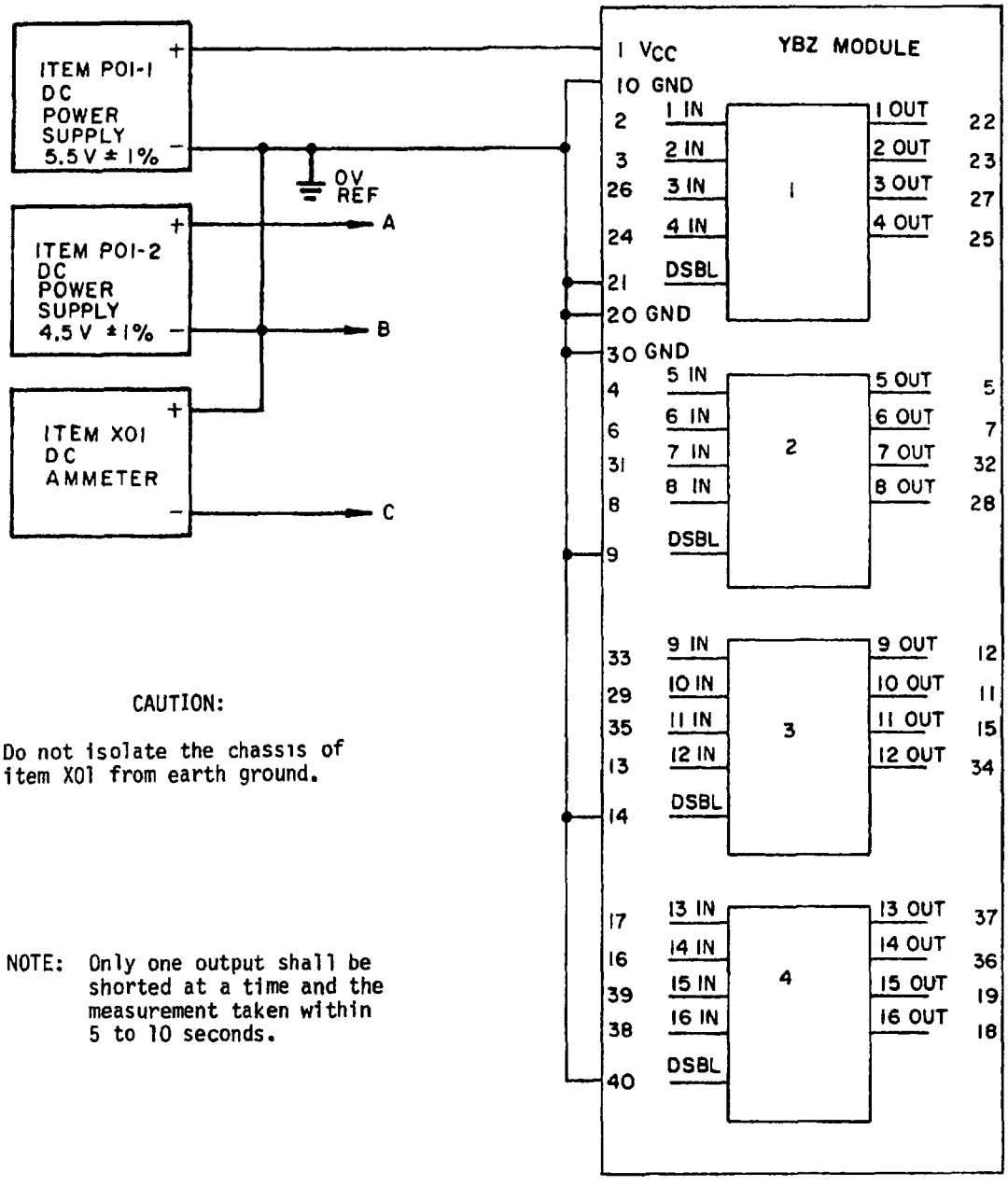
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	1 Output short circuit current	2	3,26,24,4,6,31,8,33, 29,35,13,17,16,39,38,	22	-30	-70	mA	9
E2	2 "	3	2,26,24,4,6,31,8,33, 29,35,13,17,16,39,38	23	"	"	"	"
E3	3 "	26	2,3,24,4,6,31,8,33, 29,35,13,17,16,39,38	27	"	"	"	"
E4	4 "	24	2,3,26,4,6,31,8,33, 29,35,13,17,16,39,38	25	"	"	"	"
E5	5 "	4	2,3,26,24,6,31,8,33, 29,35,13,17,16,39,38	5	"	"	"	"
E6	6 "	6	2,3,26,24,4,31,8,33, 29,35,13,17,16,39,38	7	"	"	"	"
E7	7 "	31	2,3,26,24,4,6,8,33, 29,35,13,17,16,39,38	32	"	"	"	"
E8	8 "	8	2,3,26,24,4,6,31,33, 29,35,13,17,16,39,38	28	"	"	"	"
E9	9 "	33	2,3,26,24,4,6,31,8, 29,35,13,17,16,39,38	12	"	"	"	"
E10	10 "	29	2,3,26,24,4,6,31,8, 33,35,13,17,16,39,38	11	"	"	"	"
E11	11 "	35	2,3,26,24,4,6,31,8, 33,29,13,17,16,39,38	15	"	"	"	"

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TABLE VII. Output short circuit current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E12	12 Output short circuit current	13	2,3,26,24,4,6,31,8, 33,29,35,17,16,39,38	34	-30	-70	mA	9
E13	13 "	17	2,3,26,24,4,6,31,8, 33,29,35,13,16,39,38	37	"	"	"	"
E14	14 "	16	2,3,26,24,4,6,31,8, 33,29,35,13,17,39,38	36	"	"	"	"
E15	15 "	39	2,3,26,24,4,6,31,8, 33,29,35,13,17,16,38	19	"	"	"	"
E16	16 "	38	2,3,26,24,4,6,31,8, 33,29,35,13,17,16,39	18	"	"	"	"

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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

NOTE: Only one output shall be shorted at a time and the measurement taken within 5 to 10 seconds.

FIGURE 9. Output short circuit current test circuit for table VII.

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TABLE VIII. Input and output clamp diode voltage test.

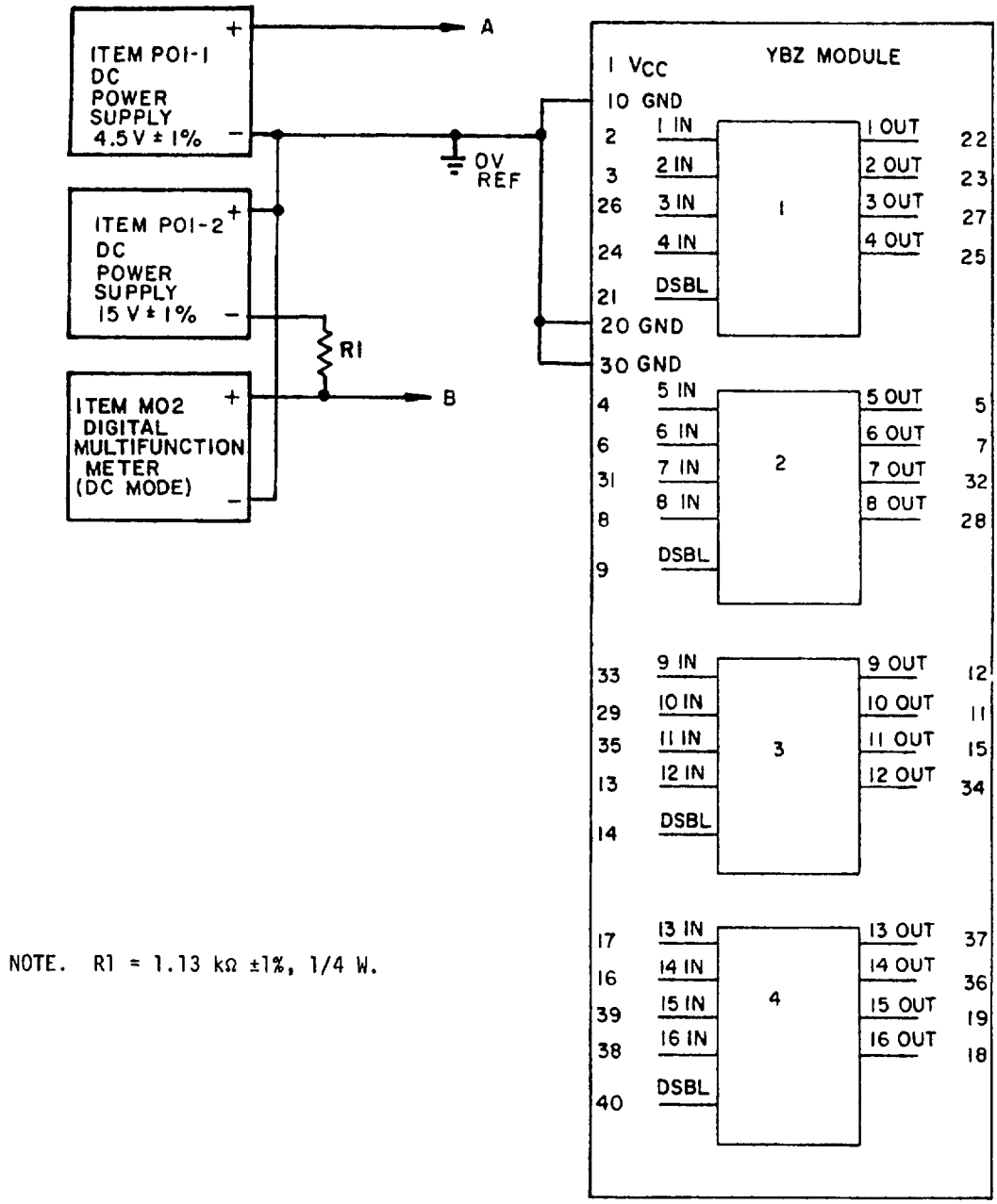
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
F1	1 Input clamp diode voltage	1	24	-0.4	-1.5	V	10
F2	2 "	"	3	"	"	"	"
F3	3 "	"	26	"	"	"	"
F4	4 "	"	24	"	"	"	"
F5	5 "	"	4	"	"	"	"
F6	6 "	"	6	"	"	"	"
F7	7 "	"	31	"	"	"	"
F8	8 "	"	8	"	"	"	"
F9	9 "	"	33	"	"	"	"
F10	10 "	"	29	"	"	"	"
F11	11 "	"	35	"	"	"	"
F12	12 "	"	13	"	"	"	"
F13	13 "	"	17	"	"	"	"
F14	14 "	"	16	"	"	"	"
F15	15 "	"	39	"	"	"	"
F16	16 "	"	38	"	"	"	"
F17	1 "	"	21	"	"	"	"
F18	2 "	"	9	"	"	"	"

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TABLE VIII. Input and output clamp diode voltage test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
F19	3 Input clamp diode voltage	1	14	-0.4	-1.5	V	10
F20	4 Input clamp diode voltage	1	40	"	"	"	"
G1	1 Output clamp diode voltage	Open	22	"	"	"	"
G2	2 "	"	23	"	"	"	"
G3	3 "	"	27	"	"	"	"
G4	4 "	"	25	"	"	"	"
G5	5 "	"	5	"	"	"	"
G6	6 "	"	7	"	"	"	"
G7	7 "	"	32	"	"	"	"
G8	8 "	"	28	"	"	"	"
G9	9 "	"	12	"	"	"	"
G10	10 "	"	11	"	"	"	"
G11	11 "	"	15	"	"	"	"
G12	12 "	"	34	"	"	"	"
G13	13 "	"	37	"	"	"	"
G14	14 "	"	36	"	"	"	"
G15	15 "	"	19	"	"	"	"
G16	16 "	"	18	"	"	"	"

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NOTE. R1 = 1.13 kΩ ±1%, 1/4 W.

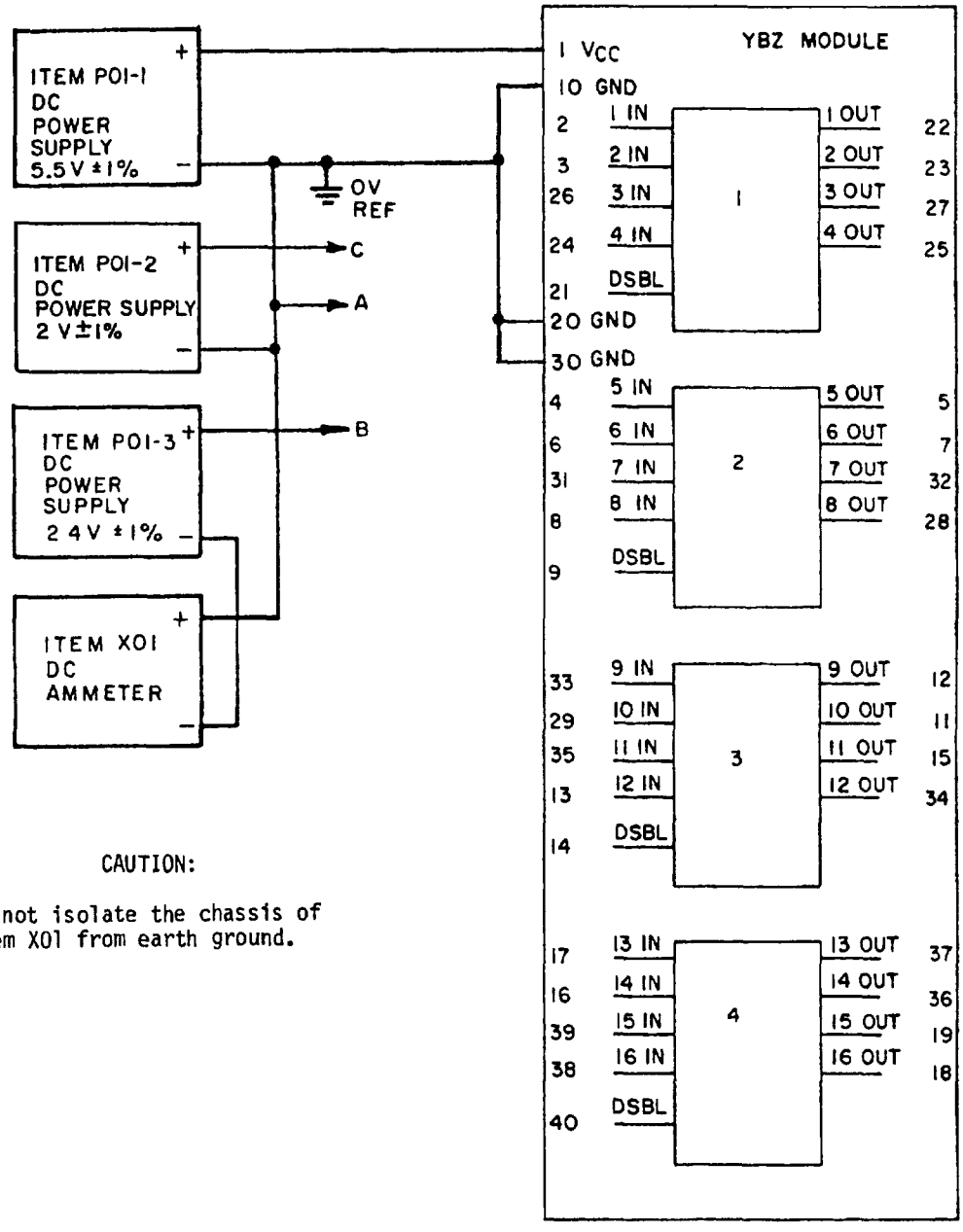
* FIGURE 10. Input and output clamp diode voltage test circuit for table VIII.

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TABLE IX. Output leakage current (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1 Out Output leakage current, high level	2	22	21	-40	40	µA	11
H2	2 " "	3	23	"	"	"	"	"
H3	3 " "	26	27	"	"	"	"	"
H4	4 " "	24	25	"	"	"	"	"
H5	5 " "	4	5	9	"	"	"	"
H6	6 " "	6	7	"	"	"	"	"
H7	7 " "	31	32	"	"	"	"	"
H8	8 " "	8	28	"	"	"	"	"
H9	9 " "	33	12	14	"	"	"	"
H10	10 " "	29	11	"	"	"	"	"
H11	11 " "	35	15	"	"	"	"	"
H12	12 " "	13	34	"	"	"	"	"
H13	13 " "	17	37	40	"	"	"	"
H14	14 " "	16	36	"	"	"	"	"
H15	15 " "	39	19	"	"	"	"	"
H16	16 " "	38	18	"	"	"	"	"

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CAUTION:

Do not isolate the chassis of item XOI from earth ground.

* FIGURE 11. Output leakage current (high-level) test circuit for table IX.

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TABLE X. Output leakage current (low-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	1 Out Output leakage current, low level	2, 21	22	-40	40	µA	12
J2	2 " "	3, 21	23	"	"	"	"
J3	3 " "	26, 21	27	"	"	"	"
J4	4 " "	24, 21	25	"	"	"	"
J5	5 " "	4, 9	5	"	"	"	"
J6	6 " "	6, 9	7	"	"	"	"
J7	7 " "	31, 9	32	"	"	"	"
J8	8 " "	8, 9	28	"	"	"	"
J9	9 " "	33, 14	12	"	"	"	"
J10	10 " "	29, 14	11	"	"	"	"
J11	11 " "	35, 14	15	"	"	"	"
J12	12 " "	13, 14	34	"	"	"	"
J13	13 " "	17, 40	37	"	"	"	"
J14	14 " "	16, 40	36	"	"	"	"
J15	15 " "	39, 40	19	"	"	"	"
J16	16 " "	40, 38	18	"	"	"	"

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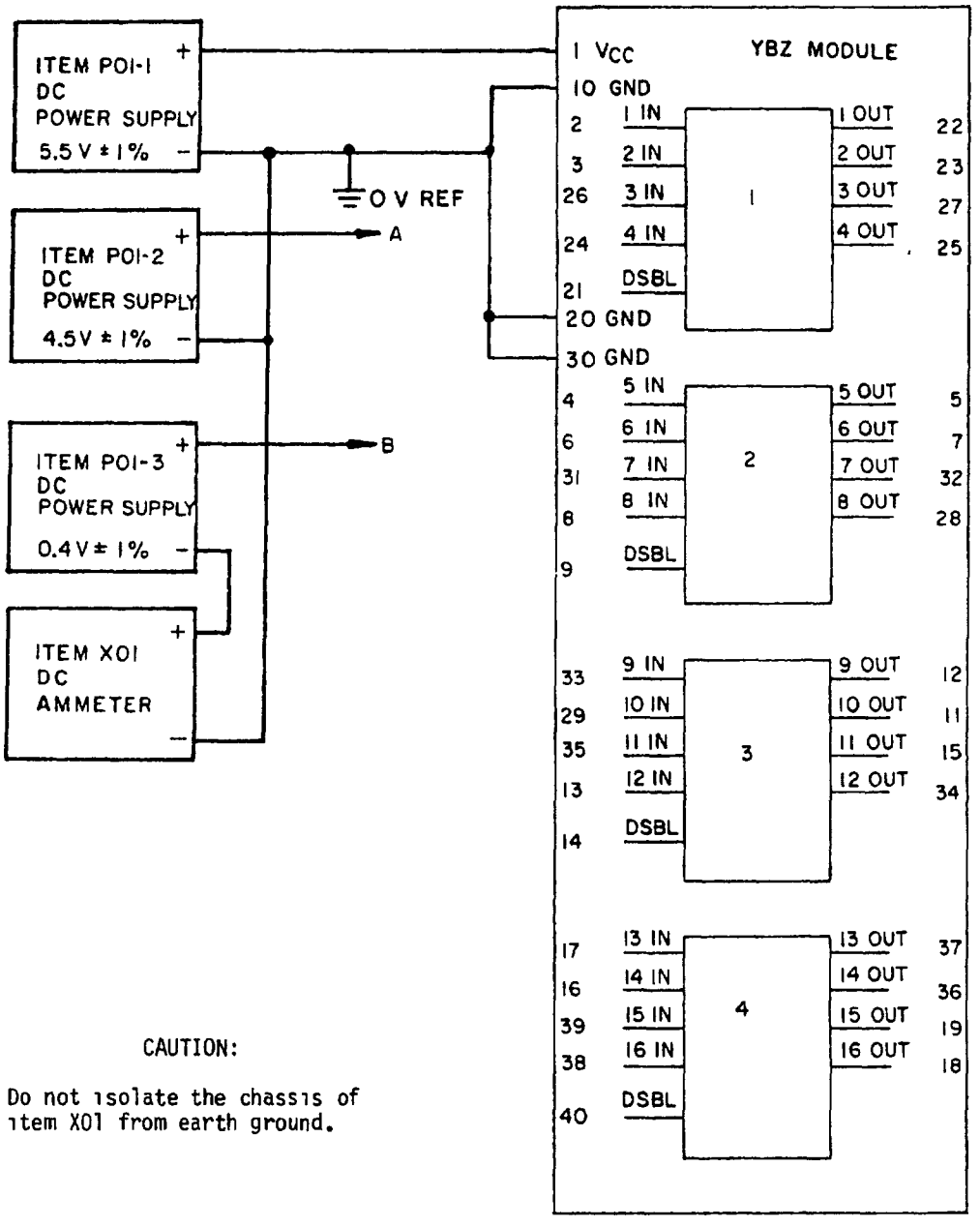


FIGURE 12. Output leakage current (low-level) test circuit for table X.

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TABLE XI. Output voltage (low-level and high-level) test.

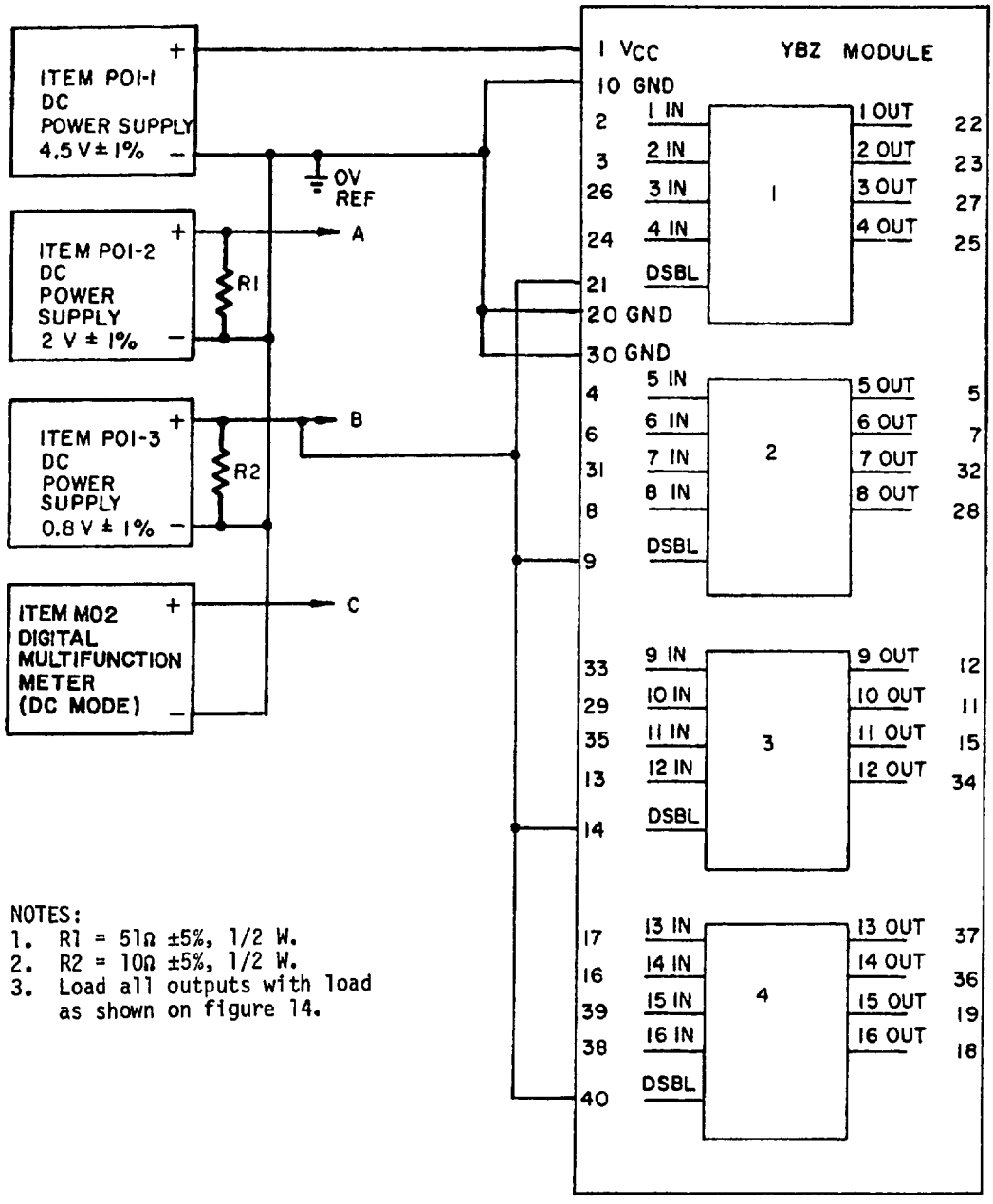
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	1 Out Output voltage, low level (I_{OL} = 16 mA)	3, 26, 24	2	22	0	400	mV	13
K2	2 " "	2, 26, 24	3	23	"	"	"	"
K3	3 " "	2, 3, 24	26	27	"	"	"	"
K4	4 " "	2, 3, 26	24	25	"	"	"	"
K5	5 " "	6, 31, 8	4	5	"	"	"	"
K6	6 " "	4, 31, 8	6	7	"	"	"	"
K7	7 " "	4, 6, 8	31	32	"	"	"	"
K8	8 " "	4, 6, 31	8	28	"	"	"	"
K9	9 " "	29, 35, 13	33	12	"	"	"	"
K10	10 " "	33, 35, 13	29	11	"	"	"	"
K11	11 " "	33, 29, 13	35	15	"	"	"	"
K12	12 " "	33, 29, 35	13	34	"	"	"	"
K13	13 " "	16, 39, 38	17	37	"	"	"	"
K14	14 " "	17, 39, 38	16	36	"	"	"	"
K15	15 " "	17, 16, 38	39	19	"	"	"	"
K16	16 " "	17, 16, 39	38	18	"	"	"	"

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TABLE XI. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	1 Out Output voltage, high level (I_{OH} = -2 mA)	2	3, 26, 24	22	2.4	4.5	V	13
L2	2 " "	3	2, 26, 24	23	"	"	"	"
L3	3 " "	26	2, 3, 24	27	"	"	"	"
L4	4 " "	24	2, 3, 26	25	"	"	"	"
L5	5 " "	4	6, 31, 8	5	"	"	"	"
L6	6 " "	6	4, 31, 8	7	"	"	"	"
L7	7 " "	31	4, 6, 8	32	"	"	"	"
L8	8 " "	8	4, 6, 31	28	"	"	"	"
L9	9 " "	33	29, 35, 13	12	"	"	"	"
L10	10 " "	29	33, 35, 13	11	"	"	"	"
L11	11 " "	35	33, 29, 13	15	"	"	"	"
L12	12 " "	13	33, 29, 35	34	"	"	"	"
L13	13 " "	17	16, 39, 38	37	"	"	"	"
L14	14 " "	16	17, 39, 38	36	"	"	"	"
L15	15 " "	39	17, 16, 38	19	"	"	"	"
L16	16 " "	38	17, 16, 39	18	"	"	"	"

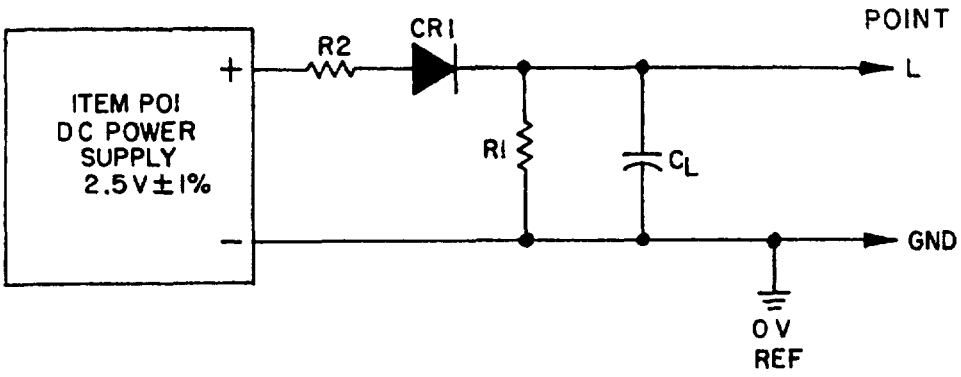
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. R2 = 10Ω ±5%, 1/2 W.
3. Load all outputs with load as shown on figure 14.

FIGURE 13. Output voltage (low-level and high-level) test circuit for table XI.

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NOTES:
1. CR1 = 1N4150, or equivalent.
2. R1 = 1.21 kΩ ±1%, 1/8 W.
3. R2 = Select value for a current flow of 16 mA ±5% out of point L with point L held at 0.4 V.
4. For test conditions Q, R, S1-S16, and T1-T16:
   C_L = 50 ±5 pF (including probe and parasitic capacitance).
5. For test conditions S17-S32 and T17-T32:
   C_L = 1,000 ±10 pF (including probe and parasitic capacitance).
6. For test conditions U, V, W, X, Y, and Z:
   C_L = 50 ±5 pF (including probe and parasitic capacitance).
7. Capacitor C_L is not required for test conditions K and L.
8. Lead length from load to module output pin under test is to be less than 1 inch (25.4 mm).

* FIGURE 14. YBZ logic load circuit.

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TABLE XII. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	1 Input current, high level	2	3,4,6,8,9,13,14,16,17,21, 24,26,29,31,33,35,38, 39,40	0	40	µA	15
M2	2 "	3	2,4,6,8,9,13,14,16,17,21, 24,26,29,31,33,35,38, 39,40	"	"	"	"
M3	3 "	26	2,3,4,6,8,9,13,14,16,17, 21,24,29,31,33,35,38,39, 40	"	"	"	"
M4	4 "	24	2,3,4,6,8,9,13,14,16,17, 21,26,29,31,33,35,38,39, 40	"	"	"	"
M5	5 "	4	2,3,6,8,9,13,14,16,17, 21,24,26,29,31,33,35,38, 39,40	"	"	"	"
M6	6 "	6	2,3,4,8,9,13,14,16,17, 21,24,26,29,31,33,35,38, 39,40	"	"	"	"
M7	7 "	31	2,3,4,6,8,9,13,14,16,17, 21,24,26,29,33,35,38,39, 40	"	"	"	"
M8	8 "	8	2,3,4,6,9,13,14,16,17, 21,24,26,39,31,33,35,38, 39,40	"	"	"	"
M9	9 "	33	2,3,4,6,8,9,13,14,16,17, 21,24,26,29,31,35,38,39, 40	"	"	"	"
M10	10 "	29	2,3,4,6,8,9,13,14,16,17, 21,24,26,31,33,35,38,39, 40	"	"	"	"

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TABLE XII. Input current (high-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
N11	11	Input current, high level	35	2,3,4,6,8,9,13,14,16,17, 21,24,26,29,31,33,38,39, 40	0	40	µA	15
N12	12	"	13	2,3,4,6,8,9,14,16,17,21, 24,26,29,31,33,35,38,39, 40	"	"	"	"
M13	13	"	17	2,3,4,6,8,9,13,14,16,21, 24,26,29,31,33,35,38,39, 40	"	"	"	"
M14	14	"	16	2,3,4,6,8,9,13,14,17,21, 24,26,29,31,33,35,38,39, 40	"	"	"	"
M15	15	"	39	2,3,4,6,8,9,13,14,16,17, 21,24,26,29,31,33,35,38, 40	"	"	"	"
M16	16	"	38	2,3,4,6,8,9,13,14,16,17, 21,24,26,29,31,33,35,39, 40	"	"	"	"
M17	Disable 1	"	21	2,3,4,6,8,9,13,14,16,17, 24,26,29,31,33,35,38,39, 40	"	160	"	"
M18	Disable 2	"	9	2,3,4,6,8,13,14,16,17, 21,24,26,29,31,33,35,38, 39,40	"	"	"	"
M19	Disable 3	"	14	2,3,4,6,8,9,13,16,17, 21,24,26,29,31,33,35,38, 39,40	"	"	"	"
M20	Disable 4	"	40	2,3,4,6,8,9,13,14,16,17, 21,24,26,29,31,33,35, 38,39	"	"	"	"

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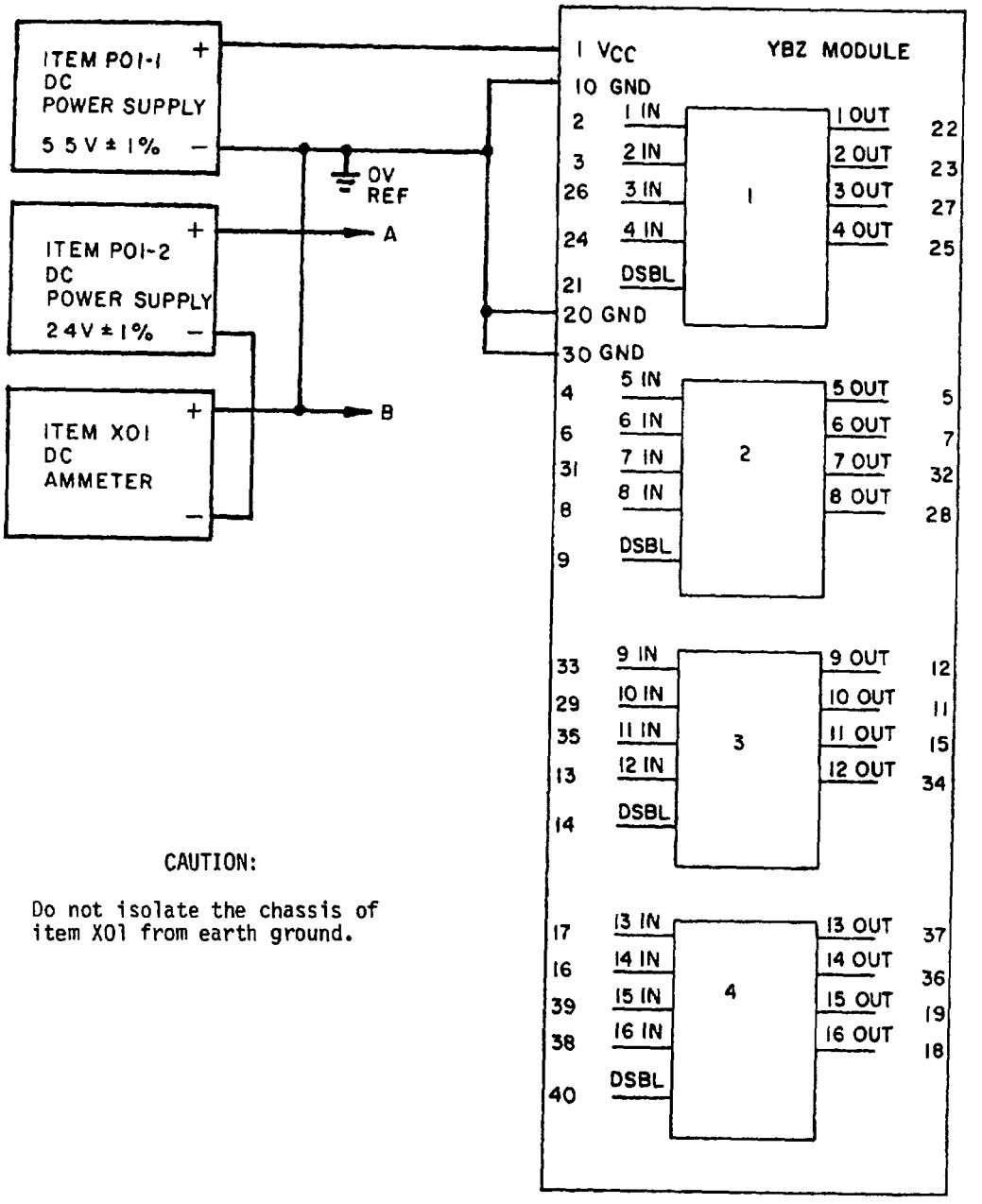


FIGURE 15. Input current (high-level) test circuit for table XII.

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TABLE XIII. Input current (low-level) test.

Test condition		Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
N1	1	Input current, low level	2	-0.10	-1.73	mA	16
N2	2	"	3	"	"	"	"
N3	3	"	26	"	"	"	"
N4	4	"	24	"	"	"	"
N5	5	"	4	"	"	"	"
N6	6	"	6	"	"	"	"
N7	7	"	31	"	"	"	"
N8	8	"	8	"	"	"	"
N9	9	"	33	"	"	"	"
N10	10	"	29	"	"	"	"
N11	11	"	35	"	"	"	"
N12	12	"	13	"	"	"	"
N13	13	"	17	"	"	"	"
N14	14	"	16	"	"	"	"
N15	15	"	39	"	"	"	"
N16	16	"	38	"	"	"	"
N17	Disable 1	"	21	"	-6.92	"	"
N18	Disable 2	"	9	"	"	"	"
N19	Disable 3	"	14	"	"	"	"
N20	Disable 4	"	40	"	"	"	"

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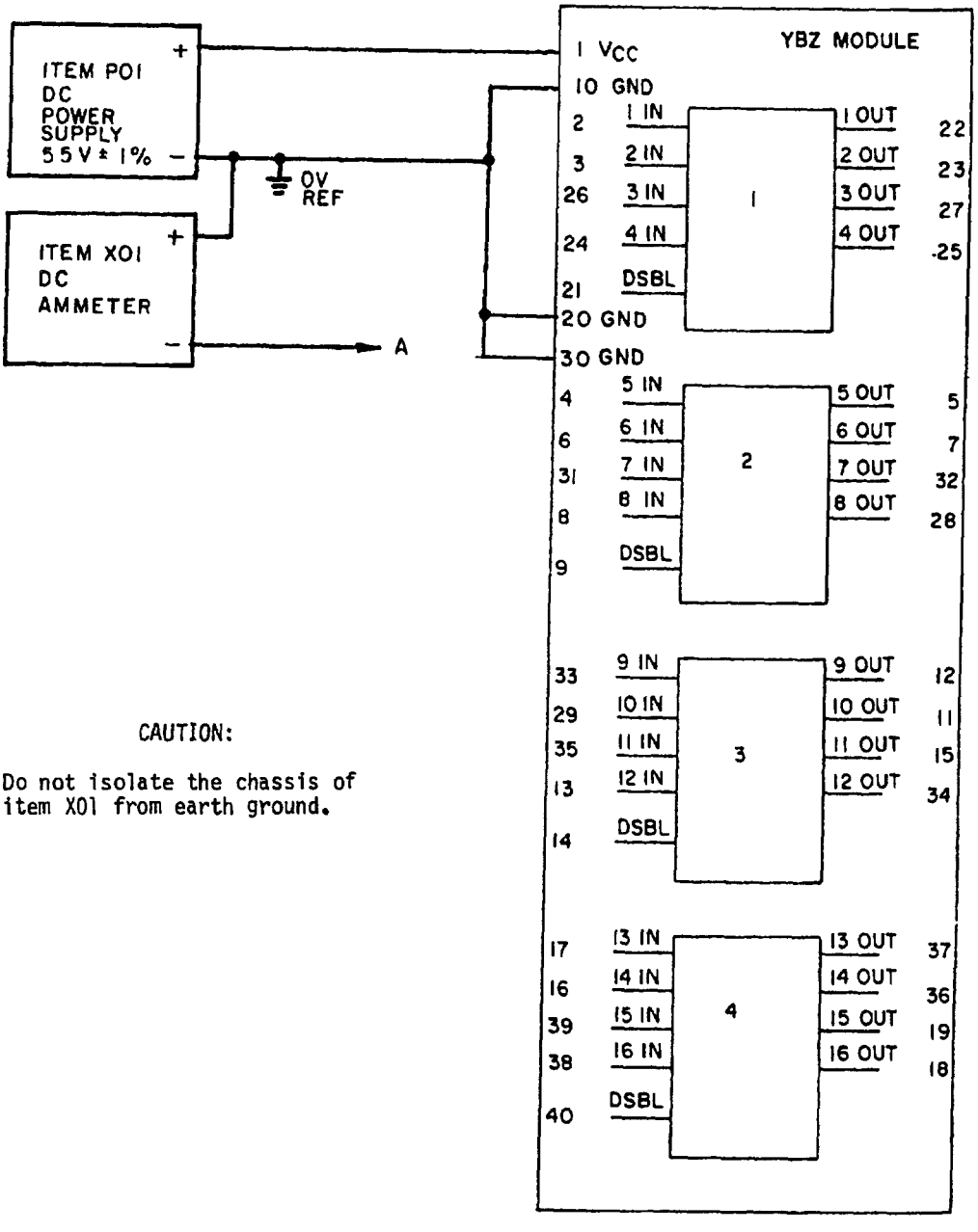


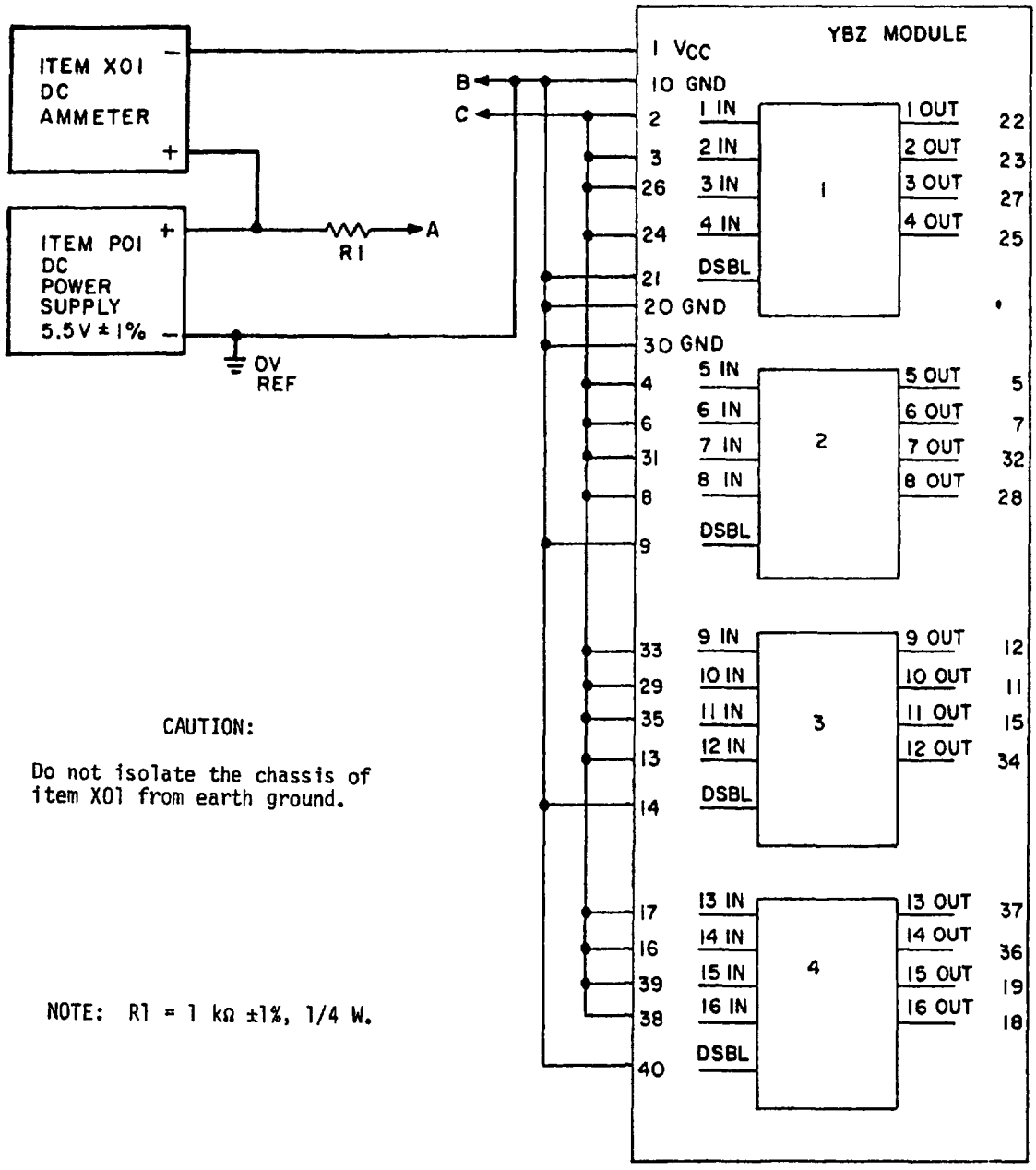
FIGURE 16. Input current (low-level) test circuit for table XIII.

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TABLE XIV. Power supply current (high-level and low-level) test.

Test condition	Parameter	Connect C to pin	Initial limits 25°C ±5°C ambient		Unit	Test figure
			Min	Max		
P1	Power supply current, high level	A	50	216	mA	17
P2	Power supply current, low level	B	50	248	mA	17

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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

NOTE: R1 = 1 kΩ ±1%, 1/4 W.

FIGURE 17. Power supply current (high-level and low-level) test circuit for table XIV.

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TABLE XV. Transition time (t_{TLH} and t_{THL}) test.

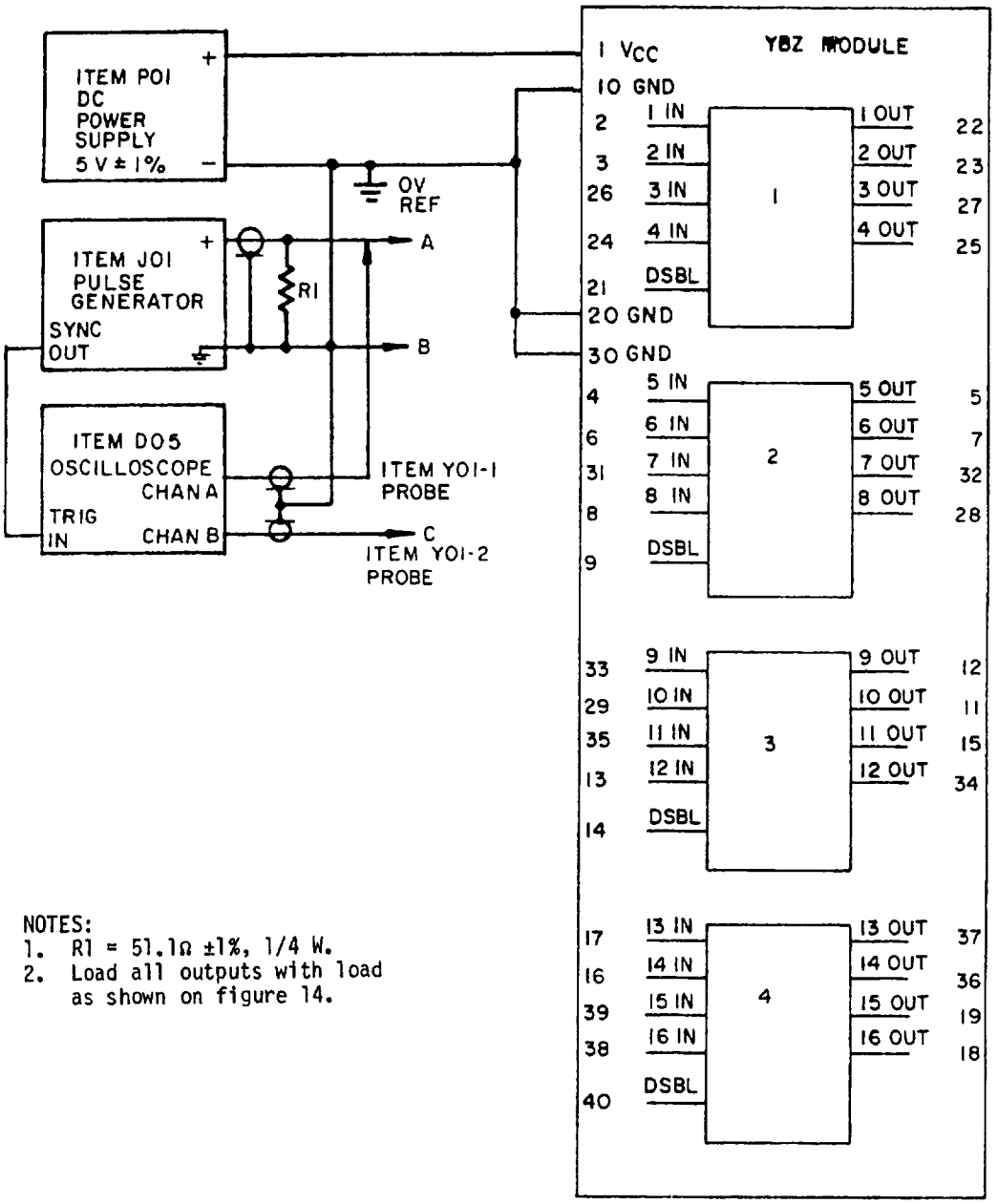
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q1	1 Out Transition time (t_{TLH}), C_L = 50 pF	2	21	22	4	25	ns	18, 19
Q2	2 " "	3	"	23	"	"	"	"
Q3	3 " "	26	"	27	"	"	"	"
Q4	4 " "	24	"	25	"	"	"	"
Q5	5 " "	4	9	5	"	"	"	"
Q6	6 " "	6	"	7	"	"	"	"
Q7	7 " "	31	"	32	"	"	"	"
Q8	8 " "	8	"	28	"	"	"	"
Q9	9 " "	33	14	12	"	"	"	"
Q10	10 " "	29	"	11	"	"	"	"
Q11	11 " "	35	"	15	"	"	"	"
Q12	12 " "	13	"	34	"	"	"	"
Q13	13 " "	17	40	37	"	"	"	"
Q14	14 " "	16	"	36	"	"	"	"
Q15	15 " "	39	"	19	"	"	"	"
Q16	16 " "	38	"	18	"	"	"	"

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TABLE XV. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R1	1 Out Transition time (t_{THL}), C_L = 50 pF	2	21	22	4	20	ns	18, 19
R2	2 " "	3	"	23	"	"	"	"
R3	3 " "	26	"	27	"	"	"	"
R4	4 " "	24	"	25	"	"	"	"
R5	5 " "	4	9	5	"	"	"	"
R6	6 " "	6	"	7	"	"	"	"
R7	7 " "	31	"	32	"	"	"	"
R8	8 " "	8	"	28	"	"	"	"
R9	9 " "	33	14	12	"	"	"	"
R10	10 " "	29	"	11	"	"	"	"
R11	11 " "	35	"	15	"	"	"	"
R12	12 " "	13	"	34	"	"	"	"
R13	13 " "	17	40	37	"	"	"	"
R14	14 " "	16	"	36	"	"	"	"
R15	15 " "	39	"	19	"	"	"	"
R16	16 " "	38	"	18	"	"	"	"

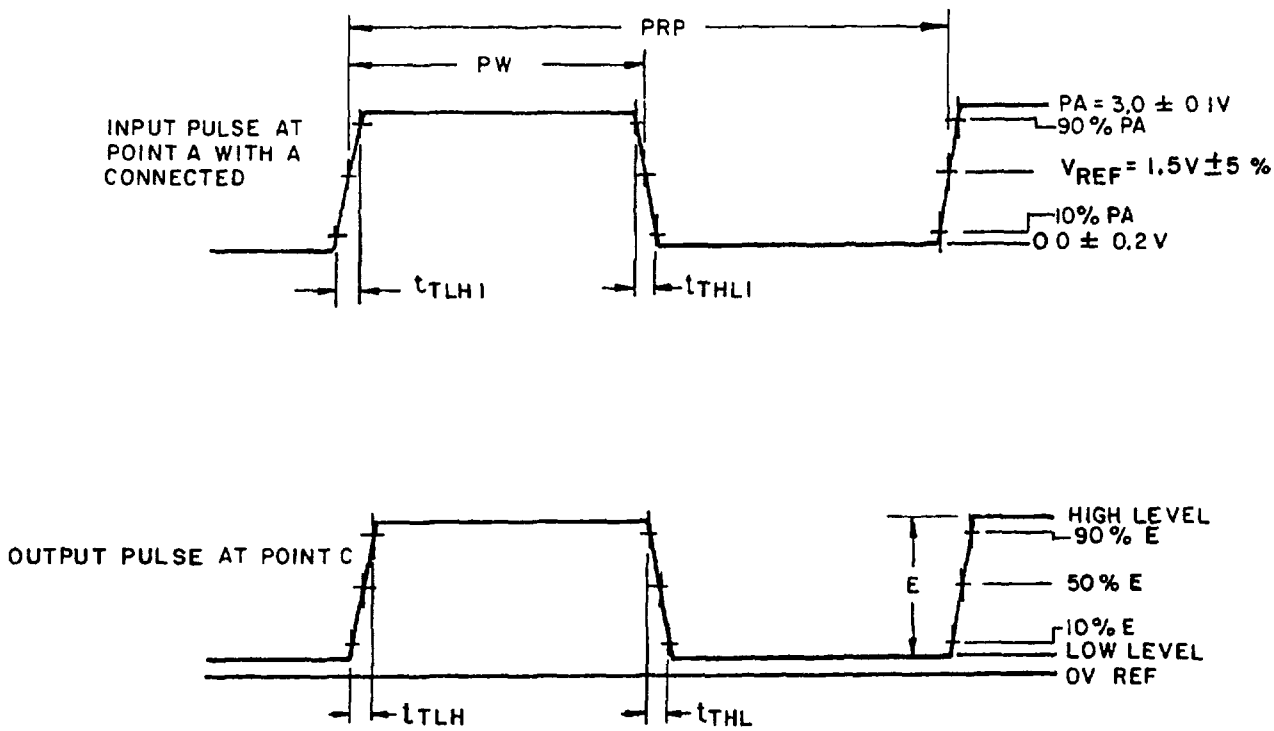
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:15Z
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Load all outputs with load as shown on figure 14.

FIGURE 18. Transition time (t_{TLH} and t_{THL}) test circuit for table XV.

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NOTES:
1. Input pulse parameters:
   PW = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
2. Before making measurements, verify that the output high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.
3. All voltage levels are referenced to ground.

* FIGURE 19. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for table XV.

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
S1	1 Propagation delay time (t_{PLH}), C_L = 50 pF	2	22	5	15	ns	20, 21
S2	2 "	3	23	"	"	"	"
S3	3 "	26	27	"	"	"	"
S4	4 "	24	25	"	"	"	"
S5	5 "	4	5	"	"	"	"
S6	6 "	6	7	"	"	"	"
S7	7 "	31	32	"	"	"	"
S8	8 "	8	28	"	"	"	"
S9	9 "	33	12	"	"	"	"
S10	10 "	29	11	"	"	"	"
S11	11 "	35	15	"	"	"	"
S12	12 "	13	34	"	"	"	"
S13	13 "	17	37	"	"	"	"
S14	14 "	16	36	"	"	"	"
S15	15 "	39	19	"	"	"	"
S16	16 "	38	18	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
S17	1 Propagation delay time (t_{PLH}), C_L = 1000 pF	2	22	10	50	ns	20, 21
S18	2 "	3	23	"	"	"	"
S19	3 "	26	27	"	"	"	"
S20	4 "	24	25	"	"	"	"
S21	5 "	4	5	"	"	"	"
S22	6 "	6	7	"	"	"	"
S23	7 "	31	32	"	"	"	"
S24	8 "	8	28	"	"	"	"
S25	9 "	33	12	"	"	"	"
S26	10 "	29	11	"	"	"	"
S27	11 "	35	15	"	"	"	"
S28	12 "	13	34	"	"	"	"
S29	13 "	17	37	"	"	"	"
S30	14 "	16	36	"	"	"	"
S31	15 "	39	19	"	"	"	"
S32	16 "	38	18	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:15Z
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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

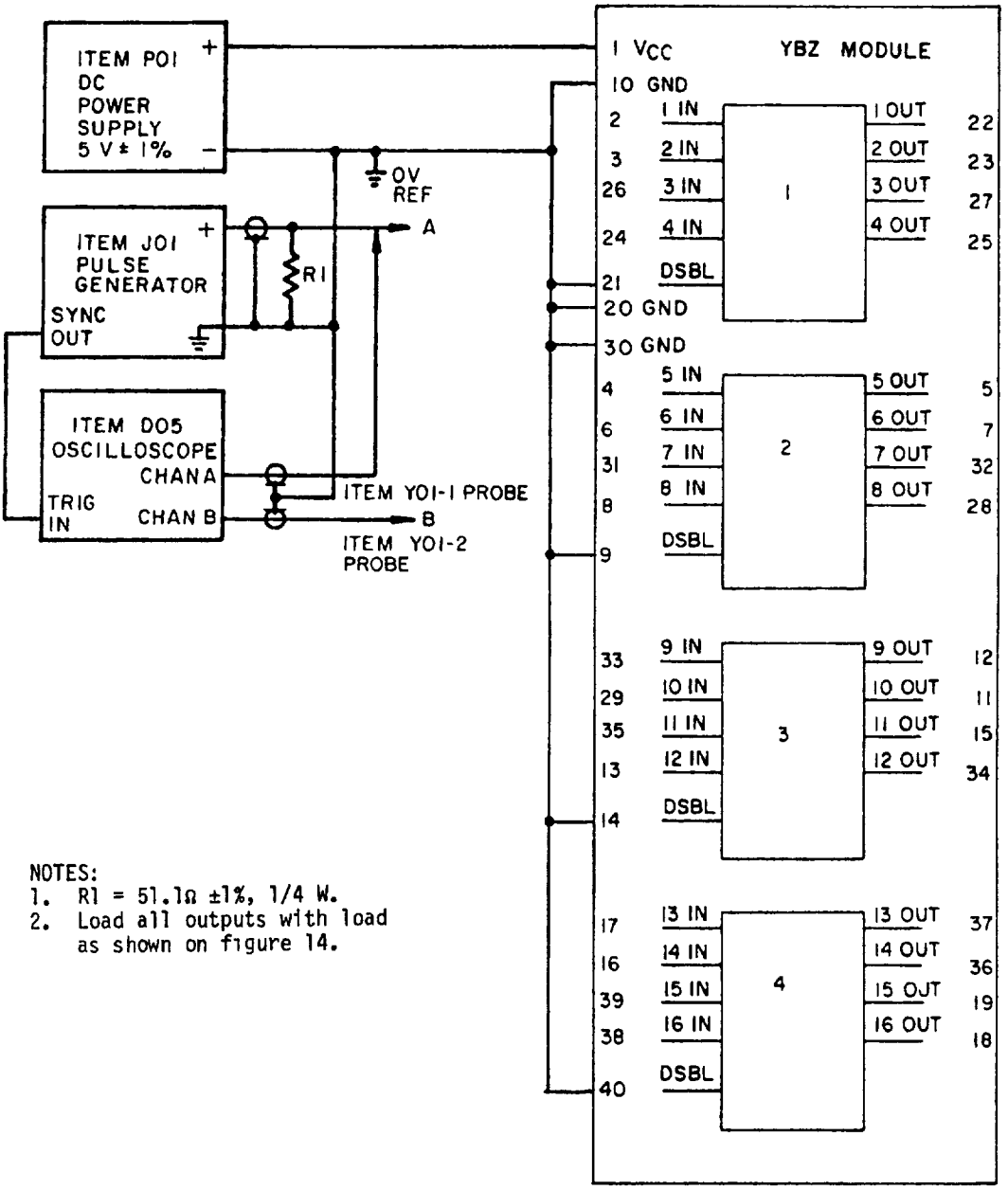
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T1	1 Propagation delay time (t_{PHL}), C_L = 50 pF	2	22	5	18	ns	20, 21
T2	2 "	3	23	"	"	"	"
T3	3 "	26	27	"	"	"	"
T4	4 "	24	25	"	"	"	"
T5	5 "	4	5	"	"	"	"
T6	6 "	6	7	"	"	"	"
T7	7 "	31	32	"	"	"	"
T8	8 "	8	28	"	"	"	"
T9	9 "	33	12	"	"	"	"
T10	10 "	29	11	"	"	"	"
T11	11 "	35	15	"	"	"	"
T12	12 "	13	34	"	"	"	"
T13	13 "	17	37	"	"	"	"
T14	14 "	16	36	"	"	"	"
T15	15 "	39	19	"	"	"	"
T16	16 "	38	18	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T17	1 Propagation delay time (t_{PHL}), C_L = 1000 pF	2	22	10	50	ns	20, 21
T18	2 "	3	23	"	"	"	"
T19	3 "	26	27	"	"	"	"
T20	4 "	24	25	"	"	"	"
T21	5 "	4	5	"	"	"	"
T22	6 "	6	7	"	"	"	"
T23	7 "	31	32	"	"	"	"
T24	8 "	8	28	"	"	"	"
T25	9 "	33	12	"	"	"	"
T26	10 "	29	11	"	"	"	"
T27	11 "	35	15	"	"	"	"
T28	12 "	13	34	"	"	"	"
T29	13 "	17	37	"	"	"	"
T30	14 "	16	36	"	"	"	"
T31	15 "	39	19	"	"	"	"
T32	16 "	38	18	"	"	"	"

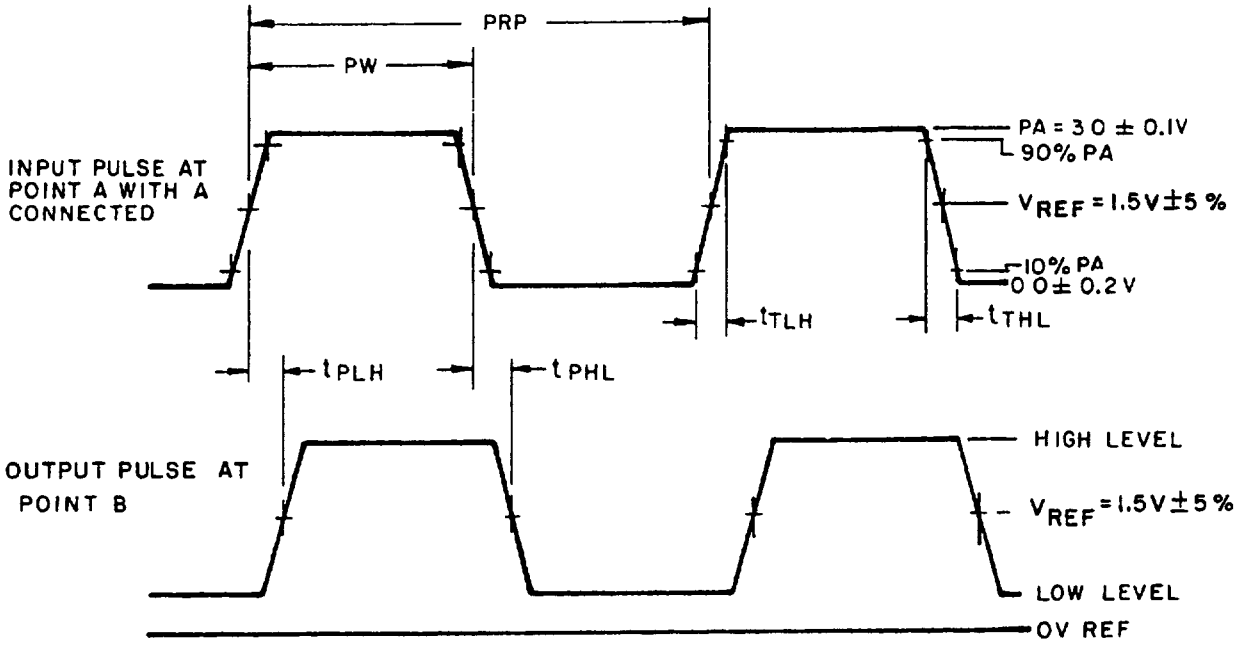
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Load all outputs with load as shown on figure 14.

FIGURE 20. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XVI.

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NOTES:
1. Input pulse parameters:
   PW = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Before making measurements, verify that the output high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.
3. All voltage levels are referenced to ground.

* FIGURE 21. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XVI.

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*TABLE XVII. Disable delay time (t_{OH}, t_{HO}, t_{1H} and t_{H1}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	1 Out Disable delay time (t_{OH}), C_L = 50 pF	Open	2,3,26,24	21	22	5	20	ns	22, 23
U2	2 " "	"	"	"	23	"	"	"	"
U3	3 " "	"	"	"	27	"	"	"	"
U4	4 " "	"	"	"	25	"	"	"	"
U5	5 " "	"	31,4,6,8,	9	5	"	"	"	"
U6	6 " "	"	"	"	7	"	"	"	"
U7	7 " "	"	"	"	32	"	"	"	"
U8	8 " "	"	"	"	28	"	"	"	"
U9	9 " "	"	13,29,33,35	14	12	"	"	"	"
U10	10 " "	"	"	"	11	"	"	"	"
U11	11 " "	"	"	"	15	"	"	"	"
U12	12 " "	"	"	"	34	"	"	"	"
U13	13 " "	"	16,17,38,39	40	37	"	"	"	"
U14	14 " "	"	"	"	36	"	"	"	"
U15	15 " "	"	"	"	19	"	"	"	"
U16	16 " "	"	"	"	18	"	"	"	"

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*TABLE XVII. Disable delay time (t_{OH}, t_{HO}, t_{1H} and t_{H1}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V1	1 Out Disable delay time (t_{HO}), C_L = 50 pF	Open	3,2,24,26	21	22	5	25	ns	22, 23
V2	2 " "	"	"	"	23	"	"	"	"
V3	3 " "	"	"	"	27	"	"	"	"
V4	4 " "	"	"	"	25	"	"	"	"
V5	5 " "	"	6,4,8,31	9	5	"	"	"	"
V6	6 " "	"	"	"	7	"	"	"	"
V7	7 " "	"	"	"	32	"	"	"	"
V8	8 " "	"	"	"	28	"	"	"	"
V9	9 " "	"	13,33,35,29	14	12	"	"	"	"
V10	10 " "	"	"	"	11	"	"	"	"
V11	11 " "	"	"	"	15	"	"	"	"
V12	12 " "	"	"	"	34	"	"	"	"
V13	13 " "	"	16,17,38,39	40	37	"	"	"	"
V14	14 " "	"	"	"	36	"	"	"	"
V15	15 " "	"	"	"	19	"	"	"	"
V16	16 " "	"	"	"	18	"	"	"	"

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*TABLE XVII. Disable delay time (t_{0H}, t_{H0}, t_{1H} and t_{H1}) test - Continued.

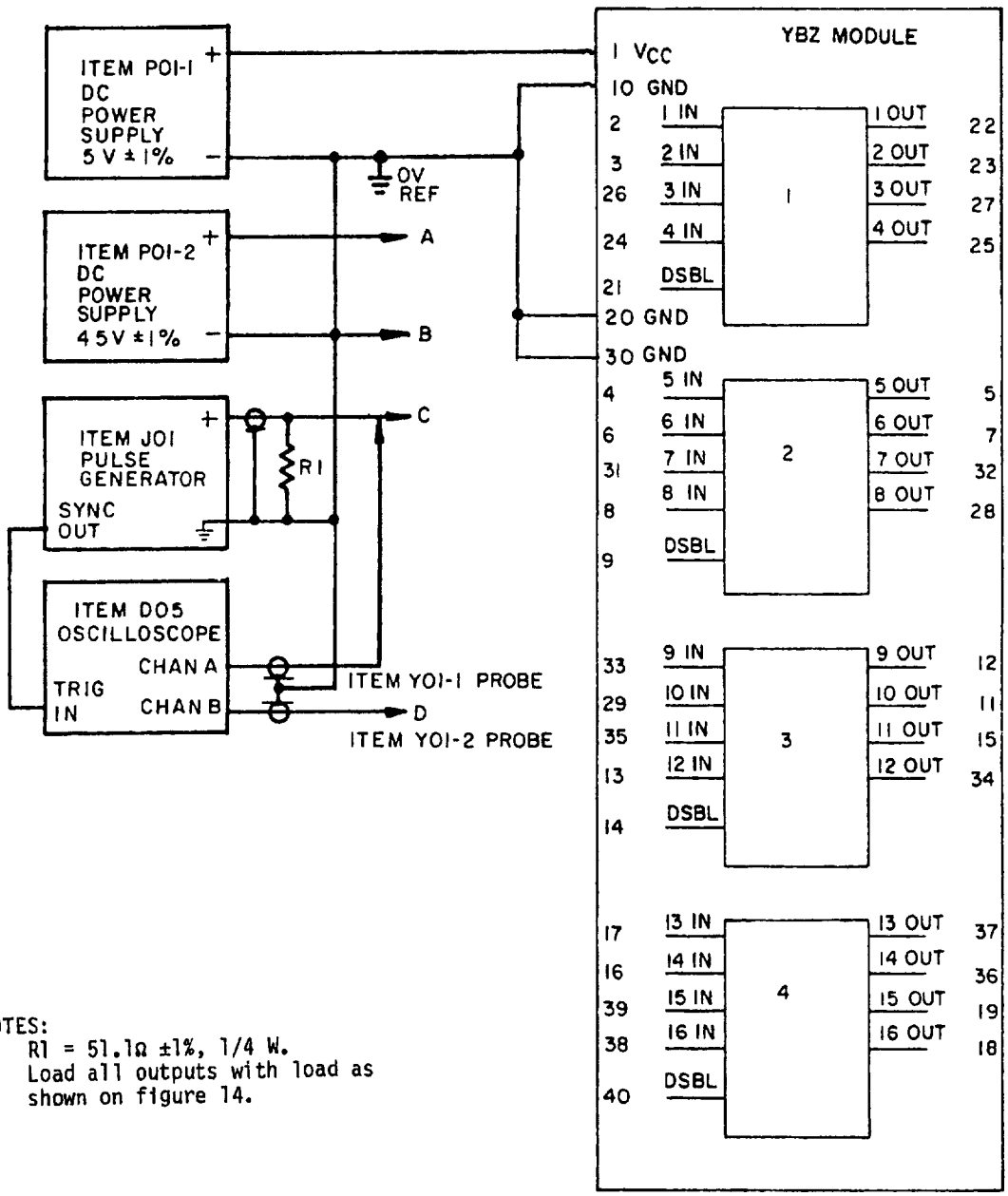
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W1	1 Out Disable delay time (t_{1H}), C_L = 50 pF	3,2,24,26	Open	21	22	5	25	ns	22, 23
W2	2 " "	"	"	"	23	"	"	"	"
W3	3 " "	"	"	"	27	"	"	"	"
W4	4 " "	"	"	"	25	"	"	"	"
W5	5 " "	6,4,31,8	"	9	5	"	"	"	"
W6	6 " "	"	"	"	7	"	"	"	"
W7	7 " "	"	"	"	32	"	"	"	"
W8	8 " "	"	"	"	28	"	"	"	"
W9	9 " "	13,33,35,29	"	14	12	"	"	"	"
W10	10 " "	"	"	"	11	"	"	"	"
W11	11 " "	"	"	"	15	"	"	"	"
W12	12 " "	"	"	"	34	"	"	"	"
W13	13 " "	16,17,38,39	"	40	37	"	"	"	"
W14	14 " "	"	"	"	36	"	"	"	"
W15	15 " "	"	"	"	19	"	"	"	"
W16	16 " "	"	"	"	18	"	"	"	"

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*TABLE XVII. Disable delay time (t_{OH}, t_{HO}, t_{1H} and t_{H1}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
X1	1 Out Disable delay time (t_{H1}), C_L = 50 pF	3,2,24,26	Open	21	22	5	23	ns	22, 23
X2	2 " "	"	"	"	23	"	"	"	"
X3	3 " "	"	"	"	27	"	"	"	"
X4	4 " "	"	"	"	25	"	"	"	"
X5	5 " "	6,4,8,31	"	9	5	"	"	"	"
X6	6 " "	"	"	"	7	"	"	"	"
X7	7 " "	"	"	"	32	"	"	"	"
X8	8 " "	"	"	"	28	"	"	"	"
X9	9 " "	13,33,35,29	"	14	12	"	"	"	"
X10	10 " "	"	"	"	11	"	"	"	"
X11	11 " "	"	"	"	15	"	"	"	"
X12	12 " "	"	"	"	34	"	"	"	"
X13	13 " "	16,17,38,39	"	40	37	"	"	"	"
X14	14 " "	"	"	"	36	"	"	"	"
X15	15 " "	"	"	"	19	"	"	"	"
X16	16 " "	"	"	"	18	"	"	"	"

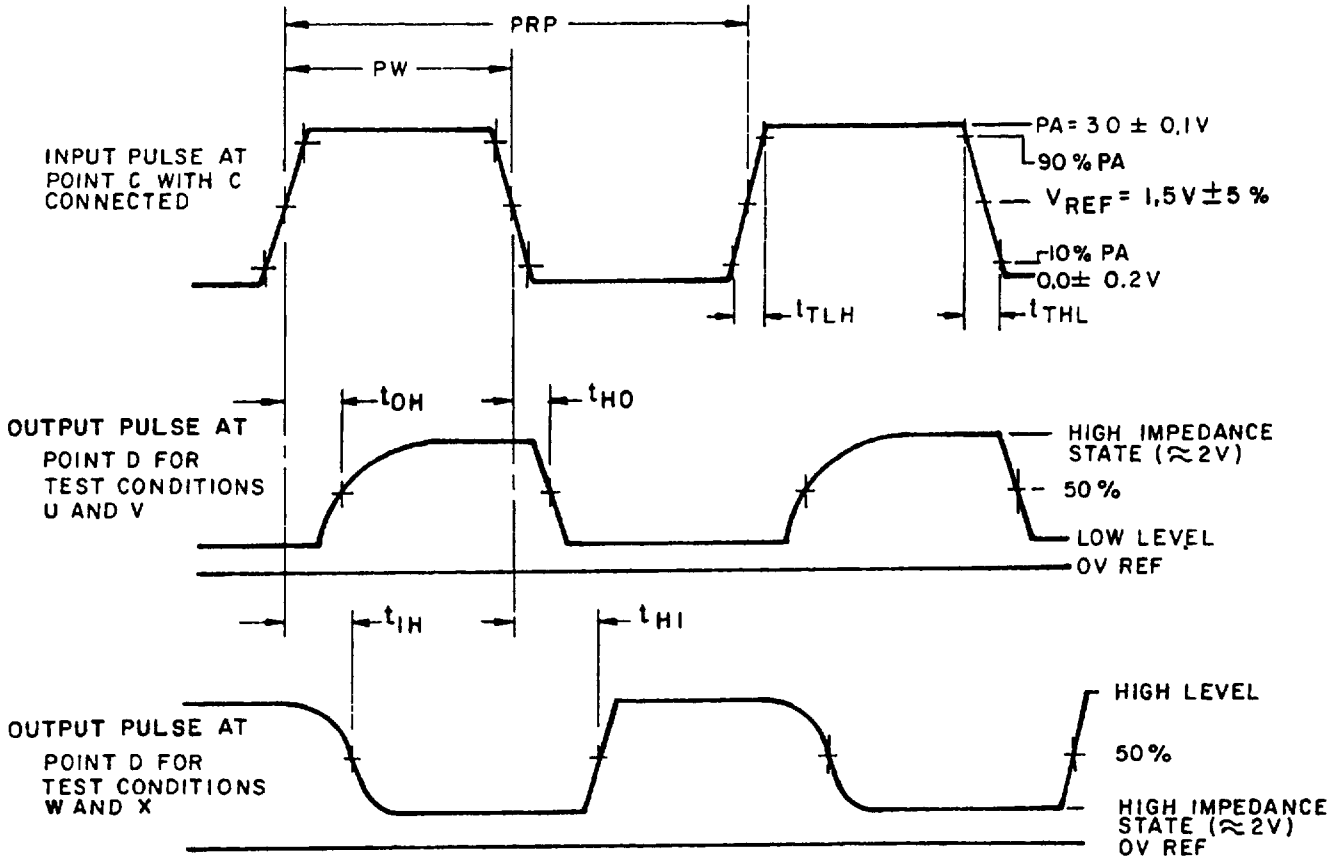
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:15Z
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Load all outputs with load as shown on figure 14.

FIGURE 22. Disable delay time (t_{0H}, t_{H0}, t_{1H}, t_{H1}) test circuit for table XVII.

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NOTES:
1. Input pulse parameters:
   PW = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Before making measurements, verify that the output high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.
3. All voltage levels are referenced to ground.

* FIGURE 23. Disable delay time (t_{OH}, t_{HO}, t_{IH}, t_{HI}) test circuit waveforms for table XVII.

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TABLE XVIII. Output crosstalk (high-level and low-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Y1	1 Out Output crosstalk, high level	2	Open	3,26,24,4,6,31,8,33,29,35,13, 17,16,39,38	22	0	400	mV	24,25
Y2	2 " "	3	"	2,26,24,4,6,31,8,33,29,35,13, 17,16,39,38	23	"	"	"	"
Y3	3 " "	26	"	2,3,24,4,6,31,8,33,29,35,13, 17,16,39,38	27	"	"	"	"
Y4	4 " "	24	"	2,3,26,4,6,31,8,33,29,35,13, 17,16,39,38	25	"	"	"	"
Y5	5 " "	4	"	2,3,26,24,6,31,8,33,29,35,13, 17,16,39,38	5	"	"	"	"
Y6	6 " "	6	"	2,3,26,24,4,31,8,33,29,35,13, 17,16,39,38	7	"	"	"	"
Y7	7 " "	31	"	2,3,26,24,4,6,8,33,29,35,13, 17,16,39,38	32	"	"	"	"
Y8	8 " "	8	"	2,3,26,24,4,6,31,33,29,35,13, 17,16,39,38	28	"	"	"	"

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TABLE XVIII. Output crosstalk (high-level and low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Y9	9 Out Output crosstalk, high level	33	Open	2,3,26,24,4,6,31,8,29,35,13,17,16,39,38	12	0	400	mV	24,25
Y10	10 " "	29	"	2,3,26,24,4,6,31,8,33,35,13,17,16,39,38	11	"	"	"	"
Y11	11 " "	35	"	2,3,26,24,4,6,31,8,33,29,13,17,16,39,38	15	"	"	"	"
Y12	12 " "	13	"	2,3,26,24,4,6,31,8,33,29,35,17,16,39,38	34	"	"	"	"
Y13	13 " "	17	"	2,3,26,24,4,6,31,8,33,29,35,13,16,39,38	37	"	"	"	"
Y14	14 " "	16	"	2,3,26,24,4,6,31,8,33,29,35,13,17,39,38	36	"	"	"	"
Y15	15 " "	39	"	2,3,26,24,4,6,31,8,33,29,35,13,17,16,38	19	"	"	"	"
Y16	16 " "	38	"	2,3,26,24,4,6,31,8,33,29,35,13,17,16,39	18	"	"	"	"

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TABLE XVIII. Output crosstalk (high-level and low-level) test - Continued.

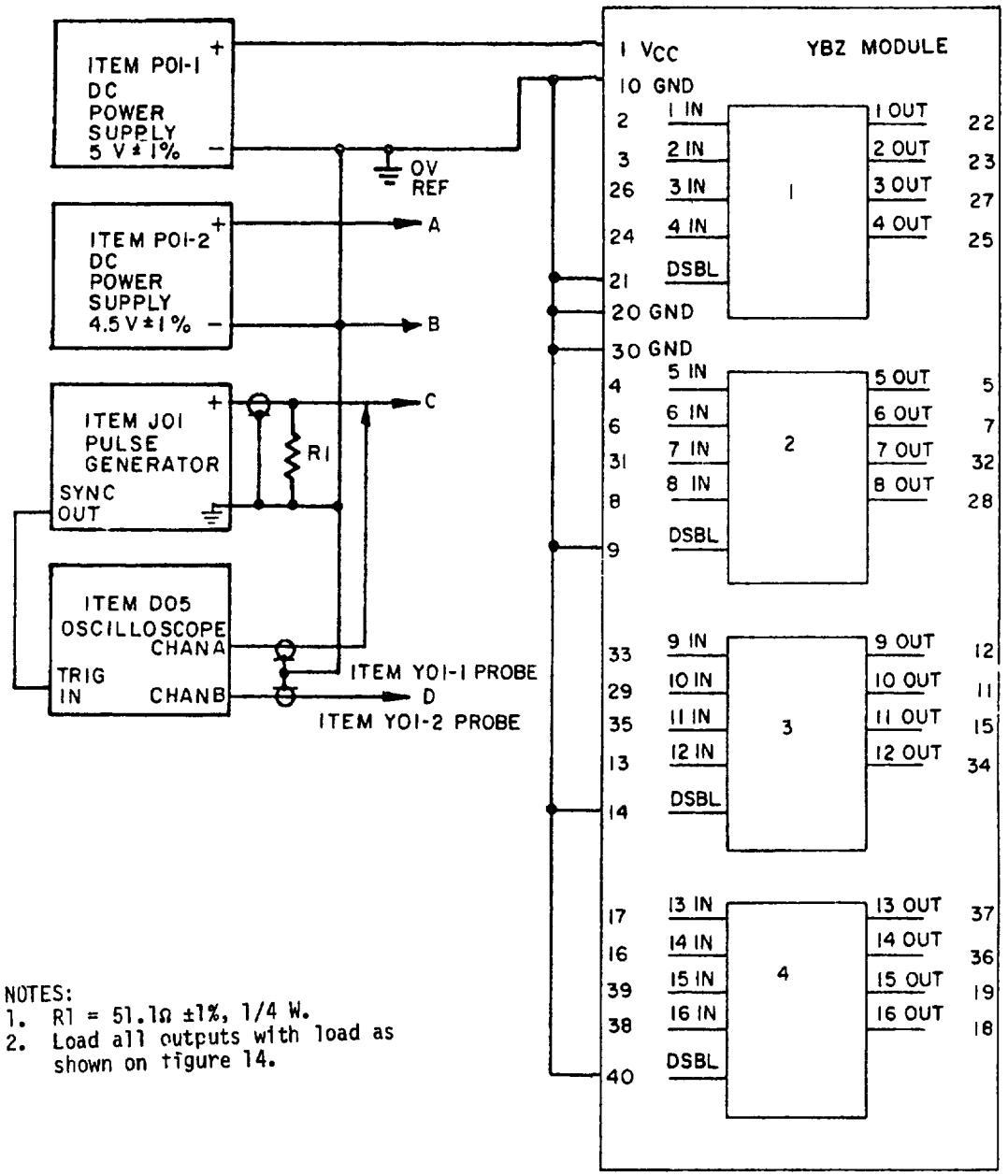
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Z1	1 Out Output crosstalk, low level	Open	2	3,26,24,4,6,31,8,33,29,35,13, 17,16,39,38	22	0	1000	mV	24,25
Z2	2 " "	"	3	2,26,24,4,6,31,8,33,29,35,13, 17,16,39,38	23	"	"	"	"
Z3	3 " "	"	26	2,3,24,4,6,31,8,33,29,35,13, 17,16,39,38	27	"	"	"	"
Z4	4 " "	"	24	2,3,26,4,6,31,8,33,29,35,13, 17,16,39,38	25	"	"	"	"
Z5	5 " "	"	4	2,3,26,24,6,31,8,33,29,35,13, 17,16,39,38	5	"	"	"	"
Z6	6 " "	"	6	2,3,26,24,4,31,8,33,29,35,13, 17,16,39,38	7	"	"	"	"
Z7	7 " "	"	31	2,3,26,24,4,6,8,33,29,35,13, 17,16,39,38	32	"	"	"	"
Z8	8 " "	"	8	2,3,26,24,4,6,31,33,29,35,13, 17,16,39,38	28	"	"	"	"

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TABLE XVIII. Output crosstalk (high-level and low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Z9	9 Out Output crosstalk, low level	Open	33	2,3,26,24,4,6,31,8,29,35,13,17,16,39,38	12	0	1000	mV	24,25
Z10	10 " "	"	29	2,3,26,24,4,6,31,8,33,35,13,17,16,39,38	11	"	"	"	"
Z11	11 " "	"	35	2,3,26,24,4,6,31,8,33,29,13,17,16,39,38	15	"	"	"	"
Z12	12 " "	"	13	2,3,26,24,4,6,31,8,33,29,35,17,16,39,38	34	"	"	"	"
Z13	13 " "	"	17	2,3,26,24,4,6,31,8,33,29,35,13,16,39,38	37	"	"	"	"
Z14	14 " "	"	16	2,3,26,24,4,6,31,8,33,29,35,13,17,39,38	36	"	"	"	"
Z15	15 " "	"	39	2,3,26,24,4,6,31,8,33,29,35,13,17,16,38	19	"	"	"	"
Z16	16 " "	"	38	2,3,26,24,4,6,31,8,33,29,35,13,17,16,39	18	"	"	"	"

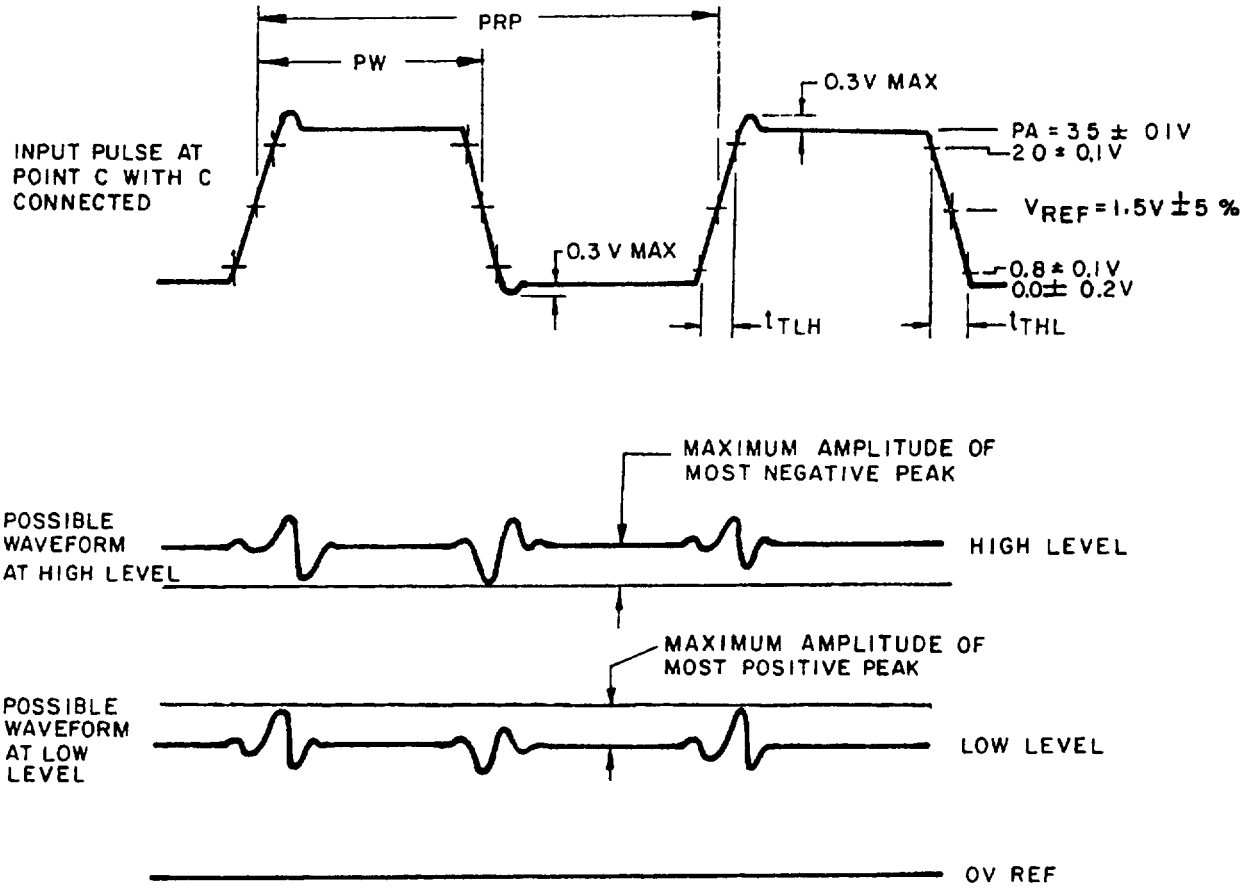
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Load all outputs with load as shown on figure 14.

FIGURE 24. Output crosstalk (high-level and low-level) test circuit for table XVIII

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NOTE: Input pulse parameters:
PW = 500 ±50 ns.
PRP = 1,000 ±100 ns.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).

FIGURE 25. Output crosstalk (high-level and low-level) test circuit waveforms for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
AA	Power supply transient amplitude	0	500	mV	26, 27

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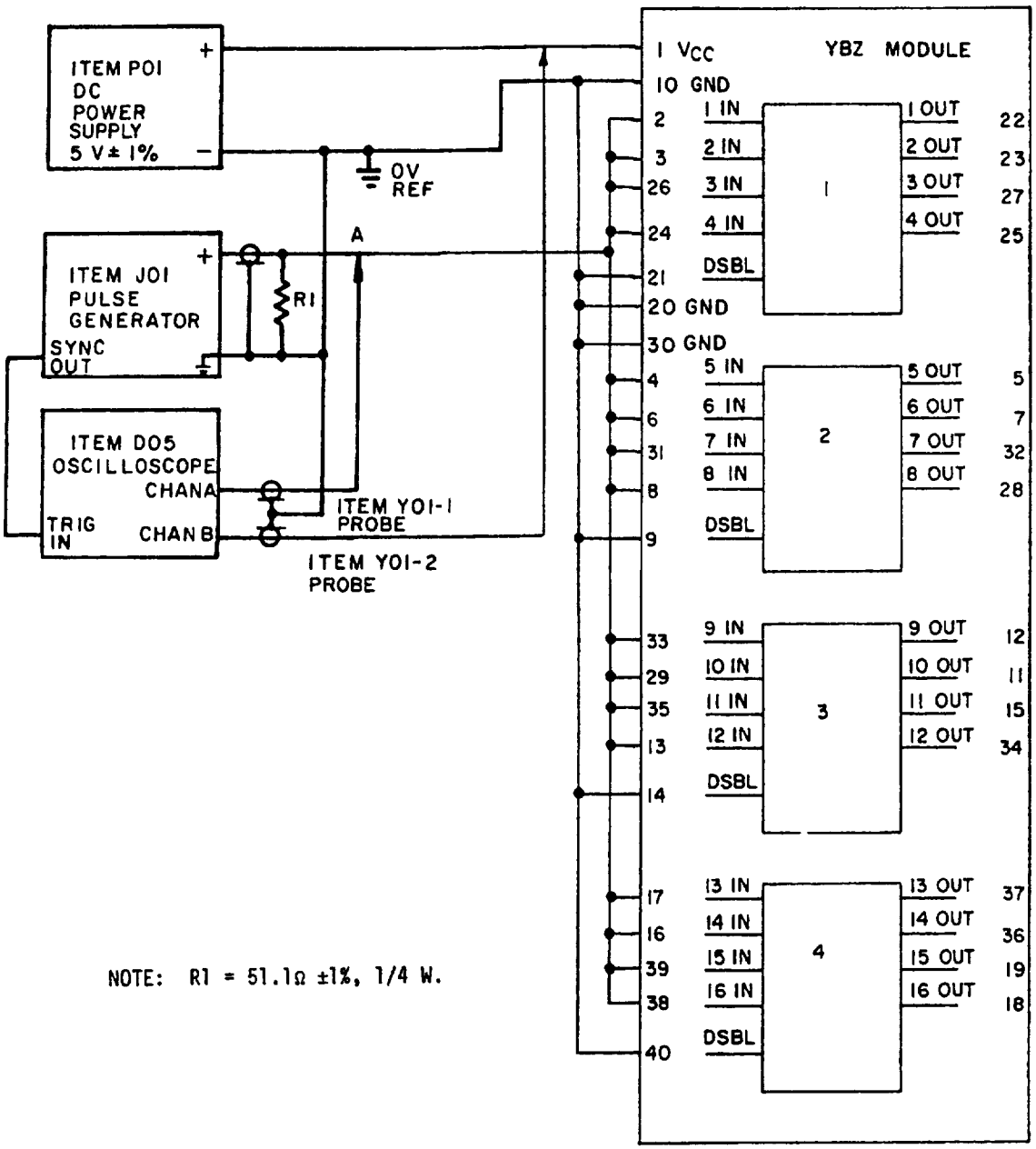
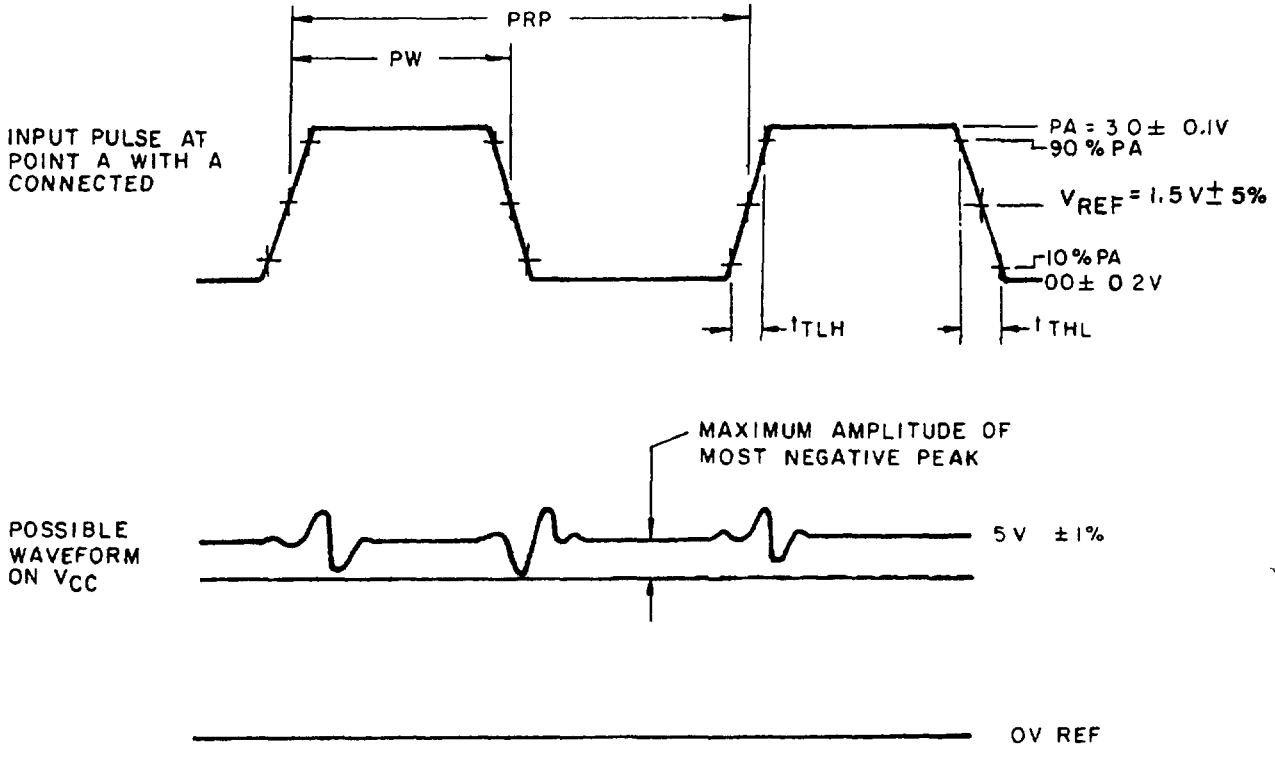


FIGURE 26. Power supply transient amplitude test circuit for table XIX.

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NOTE: Input pulse parameters:
PW = 500 ±50 ns.
PRP = 1,000 ±100 ns.
$t_{TLH} = t_{THL}$ = 10 ±2 ns.

* FIGURE 27. Power supply transient amplitude test circuit waveforms for table XIX

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TABLE XX. Power supply filter, input, and output capacitance test.

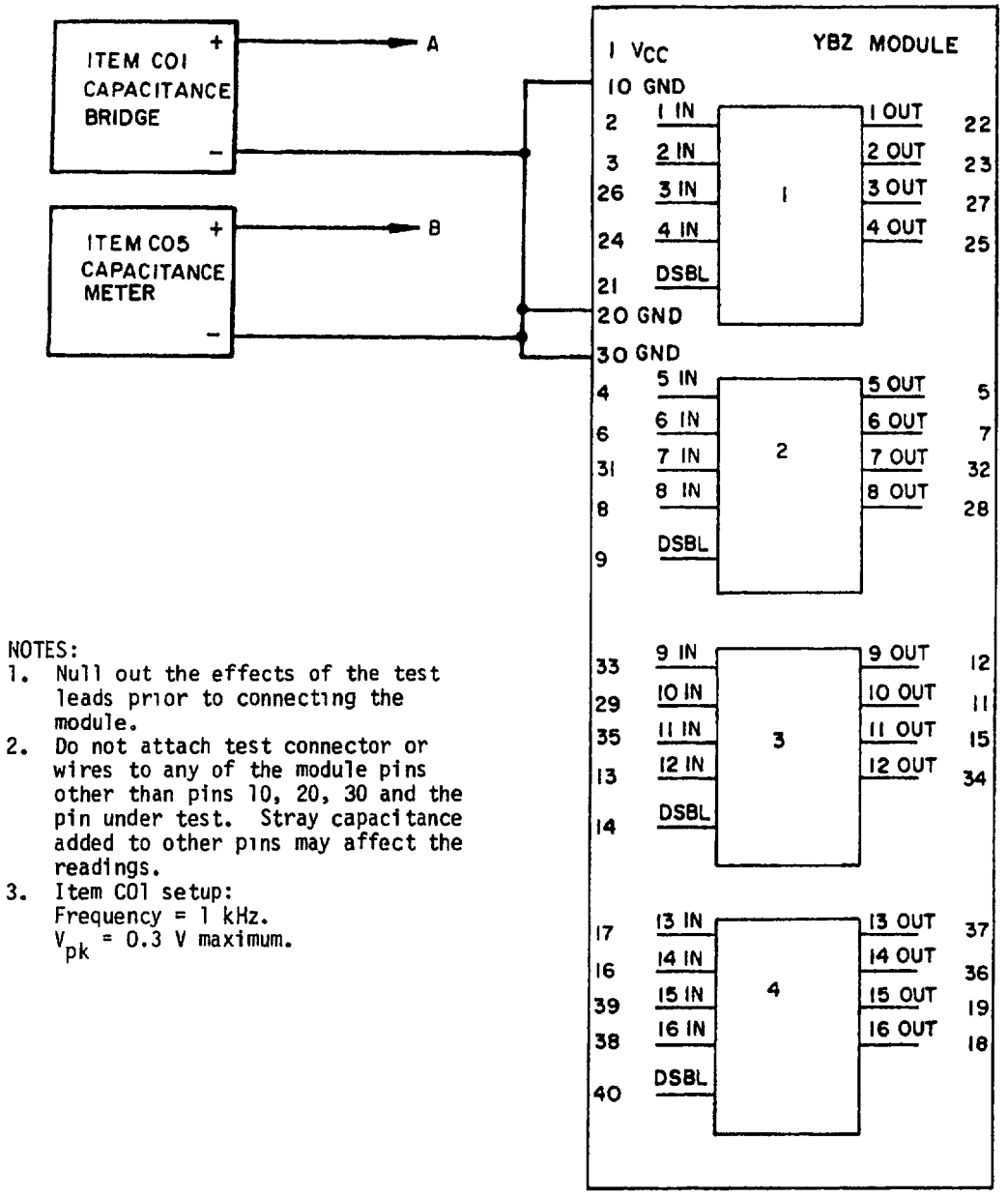
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
BB	Power supply filter capacitance	1	Open	1.6	2.8	µF	28
CC1	1 Input capacitance	Open	2	2	15	pF	"
CC2	2 "	"	3	"	"	"	"
CC3	3 "	"	26	"	"	"	"
CC4	4 "	"	24	"	"	"	"
CC5	5 "	"	4	"	"	"	"
CC6	6 "	"	6	"	"	"	"
CC7	7 "	"	31	"	"	"	"
CC8	8 "	"	8	"	"	"	"
CC9	9 "	"	33	"	"	"	"
CC10	10 "	"	29	"	"	"	"
CC11	11 "	"	35	"	"	"	"
CC12	12 "	"	13	"	"	"	"
CC13	13 "	"	17	"	"	"	"
CC14	14 "	"	16	"	"	"	"
CC15	15 "	"	39	"	"	"	"
CC16	16 "	"	38	"	"	"	"
CC17	Disable 1 "	"	21	8	60	"	"
CC18	Disable 2 "	"	9	8	60	"	"

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TABLE XX. Power supply filter, input, and output capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
CC19	Disable 3 Input capacitance	Open	14	8	60	pF	28
CC20	Disable 4 Input capacitance	"	40	8	60	"	"
DD1	1 Output capacitance	"	22	4	30	"	"
DD2	2 "	"	23	"	"	"	"
DD3	3 "	"	27	"	"	"	"
DD4	4 "	"	25	"	"	"	"
DD5	5 "	"	5	"	"	"	"
DD6	6 "	"	7	"	"	"	"
DD7	7 "	"	32	"	"	"	"
DD8	8 "	"	28	"	"	"	"
DD9	9 "	"	12	"	"	"	"
DD10	10 "	"	11	"	"	"	"
DD11	11 "	"	15	"	"	"	"
DD12	12 "	"	34	"	"	"	"
DD13	13 "	"	37	"	"	"	"
DD14	14 "	"	36	"	"	"	"
DD15	15 "	"	19	"	"	"	"
DD16	16 "	"	18	"	"	"	"

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pins 10, 20, 30 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup:
   Frequency = 1 kHz.
   V_{pk} = 0.3 V maximum.

* FIGURE 28. Power supply filter, input, and output capacitance test circuit for table XX.

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c. Reason/Rationale for Recommendation

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